Iniziato	Wednesday, 21 December 2022, 12:18
Stato	Completato
Terminato	Wednesday, 21 December 2022, 12:51
Tempo impiegato	33 min. 16 secondi
Valutazione	Non ancora valutato

Domanda **1**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the activities below is part of "Business Understanding" in the CRISP methodology?

a. Which machine learning functions are necessary for my problem?

b. Which data are available?

c. Which data must be collected with a specific campaign?

d. Which are the resources available (manpower, hardware, software, ...)

Your answer is correct.

La risposta corretta è:
Which are the resources available (manpower, hardware, software, ...)

Domanda **2**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Associate the various Data Warehouse Architectures to the respective advantages

Single Layer: The occupation of space is minimised

Two Layers: The workloads for analysis and daily operations are separated

Three Layers: An intermediate level of consolidated data is available

Risposta corretta.

La risposta corretta è:
Associate the various Data Warehouse Architectures to the respective advantages

Single Layer: [The occupation of space is minimised]

Two Layers: [The workloads for analysis and daily operations are separated]

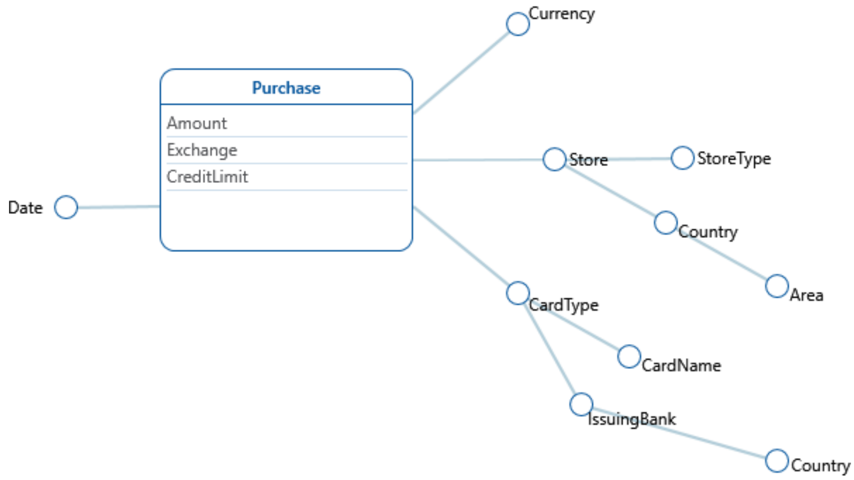
Three Layers: [An intermediate level of consolidated data is available]

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the texts describes the DFM schema below?



Scegli un'alternativa:

- a. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country. ✔
- b. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores types have also a country and an area. Purchases have a card type (e.g. credit or debit) and a card name (e.g. MyCard or JeansCard). Card names have also an issuing bank and a country.
- c. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type and country; stores have also an area. Purchases have a card name (e.g. MyCard or JeansCard), card_names have a card type (e.g. credit or debit). Card types have also an issuing bank and a country.

Risposta corretta.

La risposta corretta è: We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country.

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about ETL, which of the following activities is related to the **Extraction** step?

Scegli una o più alternative:

- [x] a. **Snapshot** of the operational data ✔
- [x] b. Association of a **timestamp** to the operational data ✔
- [ ] c. Elimination of **duplicates**
- [ ] d. Usage of dictionaries to solve **inconsistencies**

Risposta corretta.

Le risposte corrette sono: **Snapshot** of the operational data, Association of a **timestamp** to the operational data

Domanda **5**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Link the names of the OLAP operations below to their definitions

Roll-Up	Causes an increase in data aggregation and removes a detail level in a hierarchy ✔
Drill down	Reduces data aggregation and adds a detail level to a hierarchy ✔
Slice and dice	Reduces the number of cube dimensions after setting one of the dimensions to a specific value ✔
Pivot	Changes the layout, in order to analyse a group of data from a different viewpoint ✔
Drill across	Creates a link between concepts in interrelated cubes, in order to compare them ✔

Risposta corretta.

La risposta corretta è: Roll-Up → Causes an increase in data aggregation and removes a detail level in a hierarchy, Drill down → Reduces data aggregation and adds a detail level to a hierarchy, Slice and dice → Reduces the number of cube dimensions after setting one of the dimensions to a specific value, Pivot → Changes the layout, in order to analyse a group of data from a different viewpoint, Drill across → Creates a link between concepts in interrelated cubes, in order to compare them

Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about the general idea of database, what is the meaning of "Schema on write"?

Scegli un'alternativa:

- a. Create a schema for data before writing into the database ✔
- b. Create a schema for data after writing into the database
- c. Change the schema of the data after each write
- d. Create a schema for data while writing into the database

Risposta corretta.

La risposta corretta è: Create a schema for data before writing into the database

Domanda **7**

Completo

Punteggio max.: 1,00

Describe the main principles and characteristics of a Data Warehouse (not more than 300 words)

A Data Warehouse is a data collection optimised for multi-dimensional analysis, it is one of the main tools of business intelligence. A Data Warehouse takes data from different sources and always provides a unified view of them. It is non volatile, that means that data are never over written or deleted, but it is only possible to add new data and data are read-only. A DWH can be structered according to three types of architectures: a single-layer one, where the DWH is actually a virtual layer because there is an intermediate application that provides different views of the data directly from the source; A two-level architecture which separates the analysis from daily operations, so in this case data are processed according to ETL paradigm and then stored in the DWH; finally, a three-level architecture that increases the separation and adds to previous one an intermediate level of consolidated data.

Domanda **8**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Why the TF-IDF of a stop word is zero ?

- a. because the IDF is 1 ✔
- b. because the TF is 1
- c. because the IDF is 0
- d. because the TF is 0

Your answer is correct.

La risposta corretta è:
because the IDF is 1

Domanda **9**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The R-precision of a query q measures

- a. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q ✔
- b. the number of relevant documents for the query q in the first R retrieved docs, divided by the number of docs in the corpus
- c. the number of relevant documents for the query q in the first R retrieved docs, divided by R which is the number of docs retrieved
- d. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for each q

Your answer is correct.

La risposta corretta è:
the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q

Domanda **10**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

In a Chi-square test for feature selection of a term t with respect to a class c, the p-value is 0.015

- a. the term t is not selected as a feature if the confidence level is 0.99
- b. the term t is selected as a feature if the confidence level is 0.99 ✖
- c. the term t is not selected as a feature if the confidence level is 0.95
- d. the term t is selected as a feature if the confidence level is larger than 0.99

Your answer is incorrect.

La risposta corretta è:
the term t is not selected as a feature if the confidence level is 0.99

Domanda **11**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

- a. 1. onehot encoding vector of q with respect to the dictionary

  2. term weighting of the onehot encoded vector q

  3. fold-in of the term weighted vector q in the LSA space

  4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues

  5. return the top-k highest cosine similarity LSA vector terms

b.
1. onehot encoding vector of q with respect to the dictionary

2. term weighting of the onehot encoded vector q

3. fold-in of the term weighted vector q in the LSA space

4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues

5. return the top-k highest cosine similarity LSA vector terms

c.
1. onehot encoding vector of q with respect to the dictionary

2. term weighting of the onehot encoded vector q

3. fold-in of the term weighted vector q in the LSA space

4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues

5. return the top-k highest cosine similarity LSA vector documents

d.
1. onehot encoding vector of q with respect to the dictionary

2. term weighting of the onehot encoded vector q

3. fold-in of the term weighted vector q in the LSA space

4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues

5. return the top-k highest cosine similarity LSA vector documents

Your answer is incorrect.

La risposta corretta è:
1. onehot encoding vector of q with respect to the dictionary

2. term weighting of the onehot encoded vector q

3. fold-in of the term weighted vector q in the LSA space

4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues

5. return the top-k highest cosine similarity LSA vector terms

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following terms is responsible for the quadratic complexity of the attention mechanism used inside Transformers (i.e., Attention(Q, K, V) = softmax(QK^T / sqrt(d_k))V) ?

- a. The product between Q and K^T
- b. The product between softmax(QK^T) and V
- c. The factor 1/sqrt(d_k)
- d. The V matrix

Your answer is correct.

La risposta corretta è:

The product between Q and K^T

Domanda **13**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following is the correct equation of the triplet loss ?

(d(x, y) denotes the euclidean distance between x and y; a, p and n are the embeddings of the anchor, positive and negative sample, respectively)

- a. L = max(0, d(a,p) – d(a,n) + \alpha)
- b. L = max(0, d(a,p) + d(a,n) + \alpha)
- c. L = max(0, \alpha - d(a,p) + d(a,n))
- d. L = max(0, \alpha - d(a,p) - d(a,n))

Your answer is correct.

La risposta corretta è:
L = max(0, d(a,p) – d(a,n) + \alpha)

Domanda **14**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

Which are the components of a general message-passing layer in a graph neural network?

- a. Message function, Permutation Invariant Aggregator, Differentiable Function
- b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator
- c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator ✖
- d. Message function, Permutation Invariant Aggregator, Attention Mechanism

Your answer is incorrect.

La risposta corretta è:

Message function, Permutation Invariant Aggregator, Differentiable Function

Domanda **15**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The phrase "Big Data Loop" refers to

- a kind of big data processing framework which processes big data using a sequence of phases
- the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites ✔
- an algorithm designed to process big data in an iterative fashion

Your answer is correct.

La risposta corretta è:
the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites

Domanda **16**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The filter() method of DataFrame in Spark's structured API

- retrieves specified columns of a data frame for its rows which satisfy a boolean condition
- selects specified, possibly renamed, columns of a data frame
- accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition ✔

Your answer is correct.

La risposta corretta è:
accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition

Domanda **17**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's structured API, calling the join method on a data frame df, with df1 as argument

- creates a new data frame containing all columns of df and df1; the computation is legal only if df and df1 have the same number of rows
- creates a new data frame containing all rows of df and df1; the computation is legal only if df and df1 have the same number of columns
- creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df ✔

Your answer is correct.

La risposta corretta è:
creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df

Domanda **18**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The map function in MapReduce

- ○ always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs
- ○ is a predefined function which processes key-value pairs; its behaviour can be changed by passing parameters
- ● is a user defined function which is applied to every key-value pair of the dataset ✔

Your answer is correct.

Le risposte corrette sono:
always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs,

is a user defined function which is applied to every key-value pair of the dataset

Domanda **19**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the Hadoop system, the MapReduce application client

- ○ centrally computes a map function, then negotiates resource allocation with YARN for the Reduce phase
- ○ autonomously computes the resources each worker is allowed to use
- ● sends requests to YARN for resource allocation ✔

Your answer is correct.

La risposta corretta è:
sends requests to YARN for resource allocation

Domanda **20**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

A difference between an RDBMS and MapReduce is

- ○ that an RDBMS is capable of performing distributed computation
- ◉ that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses ✔
- ○ that MapReduce cannot compute any form of aggregation query on the data

Your answer is correct.

Vai a...

Quiz to Data Mining, Text Mining and Big Data Analytics - Theory [copy] ►

Domanda **21**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the structured API of Spark, pyspark.sql.Row

- ○ is method of DataFrameReader which gets one row at a time from a JSON text file
- ◉ can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame ✔
- ○ is method for reading rows from a data frame

Your answer is correct.

La risposta corretta è:
can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame

Iniziato	Wednesday, 21 December 2022, 12:18
Stato	Completato
Terminato	Wednesday, 21 December 2022, 13:00
Tempo impiegato	42 min.
Valutazione	Non ancora valutato

Domanda **1**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the activities below is part of "Business Understanding" in the CRISP methodology?

- a. Which machine learning functions are necessary for my problem?
- b. Which data are available?
- c. Which data must be collected with a specific campaign?
- d. Which are the resources available (manpower, hardware, software, ...) ✔

Your answer is correct.

La risposta corretta è:
Which are the resources available (manpower, hardware, software, ...)

Domanda **2**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following are typical data warehouse queries?

Scegli una o più alternative:

- [x] a. Which products maximize the profit? ✔
- [x] b. What is the total revenue per product category and state? ✔
- [ ] c. What is the total revenue of yesterday in shop "BO03" ?
- [ ] d. Which is the stock of product "XXX" in the "YYY" warehouse
- [x] e. What is the relationship between profits gained by products "WWW" and "ZZZ"? ✔

Risposta corretta.

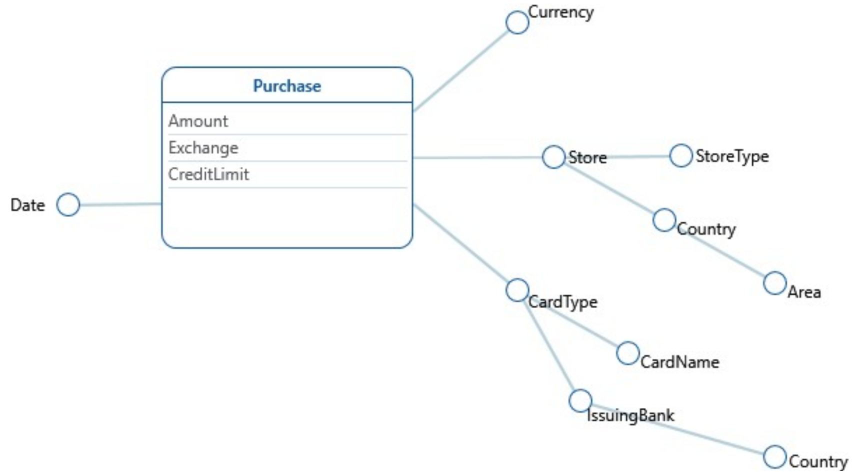
Le risposte corrette sono: Which products maximize the profit?, What is the total revenue per product category and state?, What is the relationship between profits gained by products "WWW" and "ZZZ"?

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the texts describes the DFM schema below?



Scegli un'alternativa:

- a. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country. ✔
- b. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores types have also a country and an area. Purchases have a card type (e.g. credit or debit) and a card name (e.g. MyCard or JeansCard). Card names have also an issuing bank and a country.
- c. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type and country; stores have also an area. Purchases have a card name (e.g. MyCard or JeansCard), card_names have a card type (e.g. credit or debit). Card types have also an issuing bank and a country.

Risposta corretta.

La risposta corretta è: We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country.

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about ETL, which of the following activities is related to the **Cleansing** step?

Scegli una o più alternative:

- ☐ a. **Snapshot** of the operational data
- ☐ b. Association of a **timestamp** to the operational data
- ☑ c. Elimination of **duplicates** ✔
- ☑ d. Usage of dictionaries to solve **inconsistencies** ✔

Risposta corretta.

Le risposte corrette sono: Elimination of **duplicates**, Usage of dictionaries to solve **inconsistencies**

Domanda **5**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the definition below describes the OLAP operation **Drill-Down**?

Scegli un'alternativa:

- ○ a. Creates a link between concepts in interrelated cubes, to compare them
- ○ b. Causes an increase in data aggregation and removes a detail level in a hierarchy
- ◉ c. Reduces data aggregation and adds a detail level to a hierarchy ✔
- ○ d. Reduces the number of cube dimensions after setting one of the dimensions to a specific value
- ○ e. Changes the layout, in order to analyse a group of data from a different viewpoint

Risposta corretta.

La risposta corretta è: Reduces data aggregation and adds a detail level to a hierarchy

Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about the general idea of database, what is the meaning of "Schema on write"?

Scegli un'alternativa:

- (selected) a. Create a schema for data before writing into the database ✓
- b. Create a schema for data after writing into the database
- c. Change the schema of the data after each write
- d. Create a schema for data while writing into the database

Risposta corretta.

La risposta corretta è: Create a schema for data before writing into the database

Domanda **7**

Completo

Punteggio max.: 1,00

Describe the concept of OLAP and the main differences between OLAP and OLTP. (maximum 1000 characters, the evaluation is graded according to clarity, completeness and relevance)

OLAP is the acronym for Online Analitical Processing, this is a decision support system that exploits the multidimensional data model to perform different analisys on the Data Warehouse.

There are plenty OLAP operations that analists can use to produce reports for the business:

Pivot: changes the layout of the data in order to analyze it from a different view point.

Drill-Down: reduces data aggregation by adding a more detailed field in the hierarchy.

Drill-Across: analyze relations between two different features of the data.

Roll-Up : augments data agregattion by removing a feature from data.

OLTP is the acronym for Online Transaction Processing, it is a software system that support transaction based applications, it works with relational data, acessing only few elements in a read or write mode. On the other hand OLAP operations can acess milions of records briefly but only in read mode, working only on the data found in a Data Warehouse.

Domanda **8**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Stemming and Lemmatization

- a. are text processing techniques to cut suffixes and reduce each word to its root and base form respectively
- b. are text processing techniques to cut suffixes and reduce each word to its base form and root respectively
- c. are text processing techniques to cut prefix and reduce each word to its base form and root respectively
- d. are text processing techniques to cut prefix and reduce each word to its root and base form respectively

Your answer is correct.

La risposta corretta è:
are text processing techniques to cut suffixes and reduce each word to its root and base form respectively

Domanda **9**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

The R-precision of a query q measures

- a. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q
- b. the number of relevant documents for the query q in the first R retrieved docs, divided by the number of docs in the corpus
- c. the number of relevant documents for the query q in the first R retrieved docs, divided by R which is the number of docs retrieved
- d. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for each q

Your answer is incorrect.

La risposta corretta è:
the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q

Domanda **10**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In a Chi-square test for feature selection of a term t with respect to a class c, the p-value is 0.015

- a. the term t is not selected as a feature if the confidence level is 0.99 ✔
- b. the term t is selected as a feature if the confidence level is 0.99
- c. the term t is not selected as a feature if the confidence level is 0.95
- d. the term t is selected as a feature if the confidence level is larger than 0.99

Your answer is correct.

La risposta corretta è:
the term t is not selected as a feature if the confidence level is 0.99

Domanda **11**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

a.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

b.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

c.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

d. ✖
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

Your answer is incorrect.

La risposta corretta è:

1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following terms is responsible for the quadratic complexity of the attention mechanism used inside Transformers (i.e., Attention(Q, K, V) = softmax(QK^T / sqrt(d_k))V) ?

a. The product between Q and K^T

b. The product between softmax(QK^T) and V

c. The factor 1/sqrt(d_k)

d. The V matrix

Your answer is correct.

La risposta corretta è:

The product between Q and K^T

Domanda **13**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

Which of the following is the correct equation of the triplet loss ?

(d(x, y) denotes the euclidean distance between x and y; a, p and n are the embeddings of the anchor, positive and negative sample, respectively)

- a. L = max(0, d(a,p) – d(a,n) + \alpha)
- b. L = max(0, d(a,p) + d(a,n) + \alpha)
- c. L = max(0, \alpha - d(a,p) + d(a,n)) ✖
- d. L = max(0, \alpha - d(a,p) - d(a,n))

Your answer is incorrect.

La risposta corretta è:
L = max(0, d(a,p) – d(a,n) + \alpha)

Domanda **14**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

Which are the components of a general message-passing layer in a graph neural network?

- a. Message function, Permutation Invariant Aggregator, Differentiable Function
- b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator
- c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator ✗
- d. Message function, Permutation Invariant Aggregator, Attention Mechanism

Your answer is incorrect.

La risposta corretta è:

Message function, Permutation Invariant Aggregator, Differentiable Function

Vai a...

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Quiz to Data Mining, Text Mining and Big Data Analytics - Theory [copy] ►

The phrase "Big Data Loop" refers to

- a kind of big data processing framework which processes big data using a sequence of phases
- the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites ✓
- an algorithm designed to process big data in an iterative fashion

Your answer is correct.

La risposta corretta è:
the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites

Domanda **16**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The filter() method of DataFrame in Spark's structured API

- ○ retrieves specified columns of a data frame for its rows which satisfy a boolean condition
- ○ selects specified, possibly renamed, columns of a data frame
- ◉ accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition ✔

Your answer is correct.

La risposta corretta è:
accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition

Domanda **17**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's structured API, calling the join method on a data frame df, with df1 as argument

- ○ creates a new data frame containing all columns of df and df1; the computation is legal only if df and df1 have the same number of rows
- ○ creates a new data frame containing all rows of df and df1; the computation is legal only if df and df1 have the same number of columns
- ◉ creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df ✔

Your answer is correct.

La risposta corretta è:
creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df

Domanda **18**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The map function in MapReduce

- ○ always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs
- ○ is a predefined function which processes key-value pairs; its behaviour can be changed by passing parameters
- ◉ is a user defined function which is applied to every key-value pair of the dataset ✔

Your answer is correct.

Le risposte corrette sono:
always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs,

is a user defined function which is applied to every key-value pair of the dataset

Domanda **19**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the Hadoop system, the MapReduce application client

- ○ centrally computes a map function, then negotiates resource allocation with YARN for the Reduce phase
- ○ autonomously computes the resources each worker is allowed to use
- ◉ sends requests to YARN for resource allocation ✔

Your answer is correct.

La risposta corretta è:
sends requests to YARN for resource allocation

Domanda **20**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

A difference between an RDBMS and MapReduce is

- ○ that an RDBMS is capable of performing distributed computation
- ◉ that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses ✔
- ○ that MapReduce cannot compute any form of aggregation query on the data

Your answer is correct.

La risposta corretta è:
that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses

Domanda **21**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the structured API of Spark, pyspark.sql.Row

- ○ is method of DataFrameReader which gets one row at a time from a JSON text file
- ◉ can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame ✔
- ○ is method for reading rows from a data frame

Your answer is correct.

La risposta corretta è:
can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame

Iniziato	Thursday, 19 January 2023, 15:24
Stato	Completato
Terminato	Thursday, 19 January 2023, 15:59
Tempo impiegato	35 min.
Valutazione	**21,00** su un massimo di 21,00 (**100**%)

Domanda **1**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In which part of the CRISP methodology we perform the **test design** activity?

- a. Modeling ✔
- b. Evaluation
- c. Business Understanding
- d. Data Understanding

Your answer is correct.

La risposta corretta è:
Modeling

Domanda **2**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following are typical data warehouse queries?

Scegli una o più alternative:

- [x] a. Which products maximize the profit? ✔
- [x] b. What is the total revenue per product category and state? ✔
- [ ] c. What is the total revenue of yesterday in shop "BO03" ?
- [ ] d. Which is the stock of product "XXX" in the "YYY" warehouse
- [x] e. What is the relationship between profits gained by products "WWW" and "ZZZ"? ✔

Risposta corretta.

Le risposte corrette sono: Which products maximize the profit?, What is the total revenue per product category and state?, What is the relationship between profits gained by products "WWW" and "ZZZ"?

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

What is a hierarchy in the DFM Model?

- (●) a. A directed tree connecting dimensional attributes ✔
- ( ) b. A directed tree connecting dimensional facts
- ( ) c. A tree connecting measures
- ( ) d. The inclusion of facts into more general categories

Your answer is correct.

La risposta corretta è:
A directed tree connecting dimensional attributes

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about ETL, which of the following activities is related to the **Cleansing** step?

Scegli una o più alternative:

- [ ] a. **Snapshot** of the operational data
- [ ] b. Association of a **timestamp** to the operational data
- [x] c. Elimination of **duplicates** ✔
- [x] d. Usage of dictionaries to solve **inconsistencies** ✔

Risposta corretta.

Le risposte corrette sono: Elimination of **duplicates**, Usage of dictionaries to solve **inconsistencies**

Domanda **5**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the definition below describes the OLAP operation **Drill-Down**?

Scegli un'alternativa:

- ○ a. Creates a link between concepts in interrelated cubes, to compare them
- ○ b. Causes an increase in data aggregation and removes a detail level in a hierarchy
- ◉ c. Reduces data aggregation and adds a detail level to a hierarchy ✔
- ○ d. Reduces the number of cube dimensions after setting one of the dimensions to a specific value
- ○ e. Changes the layout, in order to analyse a group of data from a different viewpoint

Risposta corretta.

La risposta corretta è: Reduces data aggregation and adds a detail level to a hierarchy

Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about the general idea of database, what is the purpose of the "Schema on read" strategy?

Scegli una o più alternative:

- [x] a. Possibility to extract data in various shapes ✔
- [ ] b. Optimisation for various types of queries
- [x] c. Flexibility for any kind of query ✔
- [x] d. Avoid preprocessing of data before writing ✔

Risposta corretta.

Le risposte corrette sono: Possibility to extract data in various shapes, Flexibility for any kind of query, Avoid preprocessing of data before writing

Domanda **7**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Check the main objectives pursued when choosing a Data Lake architecture

- [x] a. Store the data directly from the sources, as they are; the most appropriate structure will be decided later, according to the user needs ✔
- [ ] b. The data must be accurately structured, in order to provide fast answers to complex queries
- [x] c. The storage must be scalable and cheap, at the expenses of latency ✔
- [ ] d. Ensure high quality of data through an accurate cleaning and transformation step

Your answer is correct.

Le risposte corrette sono:
Store the data directly from the sources, as they are; the most appropriate structure will be decided later, according to the user needs,

The storage must be scalable and cheap, at the expenses of latency

Domanda **8**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Stemming and Lemmatization

- a. are text processing techniques to cut suffixes and reduce each word to its root and base form respectively
- b. are text processing techniques to cut suffixes and reduce each word to its base form and root respectively
- c. are text processing techniques to cut prefix and reduce each word to its base form and root respectively
- d. are text processing techniques to cut prefix and reduce each word to its root and base form respectively

Your answer is correct.

La risposta corretta è:
are text processing techniques to cut suffixes and reduce each word to its root and base form respectively

Domanda **9**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Breakeven Point of Precision-Recall is

- a. an interpolated value of a precision-recall graph where precision and recall are identical
- b. an interpolated value of a precision-recall graph where R-precision and recall are identical
- c. a value of a precision-recall graph where precision and recall are identical
- d. a value of a precision-recall graph where R-precision and recall are identical

Your answer is correct.

La risposta corretta è:
an interpolated value of a precision-recall graph where precision and recall are identical

Domanda **10**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In a Chi-square test for feature selection of a term t with respect to a class c, the p-value is 0.015

- a. the term t is not selected as a feature if the confidence level is 0.99
- b. the term t is selected as a feature if the confidence level is 0.99
- c. the term t is not selected as a feature if the confidence level is 0.95
- d. the term t is selected as a feature if the confidence level is larger than 0.99

Your answer is correct.

La risposta corretta è:
the term t is not selected as a feature if the confidence level is 0.99

Domanda **11**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

a.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

b.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

c.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

d.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

Your answer is correct.

La risposta corretta è:
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

Domanda **12**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Which is the correct sequence of layers in a typical Transformer Encoder (without considering residual connections and normalization layers) ?

a. Input embeddings -> Position Embeddings -> MultiHead Attention -> Feed Forward ✔
b. Feed Forward -> Input embeddings -> Position Embeddings -> MultiHead Attention
c. Position Embeddings -> Input embeddings -> MultiHead Attention -> Feed Forward
d. Input embeddings -> Position Embeddings -> Feed Forward -> MultiHead Attention

Your answer is correct.
La risposta corretta è:
Input embeddings -> Position Embeddings -> MultiHead Attention -> Feed Forward

Domanda **13**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

A big e-commerce company wants an information retrieval system to search for product images by providing a text query in natural language. They want their system to be as fast as possible. Which of the following is a sensible approach to address this task ?

- a. We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index) ✔
- b. We train a text encoder and an image encoder with metric learning to have a common latent space for text and images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird)
- c. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)
- d. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird)

Your answer is correct.

La risposta corretta è:

We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

Domanda **14**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which are the components of a general message-passing layer in a graph neural network?

- a. Message function, Permutation Invariant Aggregator, Differentiable Function ✔
- b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator
- c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator
- d. Message function, Permutation Invariant Aggregator, Attention Mechanism

Your answer is correct.

La risposta corretta è:

Message function, Permutation Invariant Aggregator, Differentiable Function

Domanda **15**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The phrase "Big Data Loop" refers to

- a kind of big data processing framework which processes big data using a sequence of phases
- a looping mechanism by which web site data generation accelerates as a consequence of applying analytics to web site data ✔
- an algorithm designed to process big data in an iterative fashion

Your answer is correct.

La risposta corretta è:
a looping mechanism by which web site data generation accelerates as a consequence of applying analytics to web site data

Domanda **16**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

A difference between High performance computing (HPC) and MapReduce is

- the higher number of computing units which can be used in MapReduce with respect to HPC
- the maximum size of problems which can be solved using MapReduce
- the reduced amount of programming effort required to program an applicaton in MapReduce with respect to HPC, if the problem admits a solution in terms MapReduce computing primitives ✔

Your answer is correct.

La risposta corretta è:
the reduced amount of programming effort required to program an applicaton in MapReduce with respect to HPC, if the problem admits a solution in terms MapReduce computing primitives

Domanda **17**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the Hadoop ecosystem, YARN

- is a system which comprises components both for the management of resources and for the management of process containers ✔
- runs a varying number of resource managers in the cluster, depending on the application
- communicates with the application client using a node manager

Your answer is correct.

La risposta corretta è:
is a system which comprises components both for the management of resources and for the management of process containers

Domanda **18**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

If df is a Spark DataFrame, df.selectExpr() is a DataFrame method that

- ○ chooses an expression to compute among the expressions passed to the method as arguments
- ○ creates a data frame consisting of every row of df satisfying the expression passed to the method as argument
- ◉ creates a data frame such that each of its rows is computed as the values of the sequence of the expressions passed to the method as arguments ✔

Your answer is correct.

La risposta corretta è:
creates a data frame such that each of its rows is computed as the values of the sequence of the expressions passed to the method as arguments

Domanda **19**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The outer join supported by the structured API of Spark

- ◉ may output rows with null values for all the columns of one of the joined tables ✔
- ○ always outputs null values for all the columns of the joined table having the least number of rows
- ○ always outputs rows with null values for all the columns of at least one of the joined tables

Your answer is correct.

La risposta corretta è:
may output rows with null values for all the columns of one of the joined tables

Domanda **20**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the structured API of Spark, the DataFrame class has a method where(). Executing

```
df.where(arg)
```

where df is DataFrame object, returns a DataFrame object

- (selected) with every row of df, except the ones that do not verify expression arg ✔
- with exactly the rows of df which are also in the data frame passed as arg
- with exactly the rows of df equalling the row passed as arg

Your answer is correct.

La risposta corretta è:
with every row of df, except the ones that do not verify expression arg

Domanda **21**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's MLlib, transform() is a function that

- (selected) can be used to compute predictions from a machine learning model ✔
- standardises the dataset to have mean 0 and variance 1 in all its numeric columns
- is only used to compute predictions from a machine learning model

Your answer is correct.

La risposta corretta è:
can be used to compute predictions from a machine learning model

◄ Quiz to Data Mining, Text Mining and Big Data Analytics - Theory

Vai a...

Mock Exam ►

Iniziato	Thursday, 19 January 2023, 15:24
Stato	Completato
Terminato	Thursday, 19 January 2023, 16:02
Tempo impiegato	37 min. 48 secondi
Valutazione	**19,00** su un massimo di 21,00 (**90**%)

Domanda **1**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the activities below is part of "Business Understanding" in the CRISP methodology?

- a. Which machine learning functions are necessary for my problem?
- b. Which data are available?
- c. Which data must be collected with a specific campaign?
- d. Which are the resources available (manpower, hardware, software, ...)

Your answer is correct.

La risposta corretta è:
Which are the resources available (manpower, hardware, software, ...)

Domanda **2**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following are typical data warehouse queries?

Scegli una o più alternative:

- [x] a. Which products maximize the profit? ✔
- [x] b. What is the total revenue per product category and state? ✔
- [ ] c. What is the total revenue of yesterday in shop "BO03" ?
- [ ] d. Which is the stock of product "XXX" in the "YYY" warehouse
- [x] e. What is the relationship between profits gained by products "WWW" and "ZZZ"? ✔

Risposta corretta.

Le risposte corrette sono: Which products maximize the profit?, What is the total revenue per product category and state?, What is the relationship between profits gained by products "WWW" and "ZZZ"?

Domanda **3**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

In the DFM model, what is the meaning of *aggregation?*

- ○ a. Change the view on the data moving towards the higher levels of a hierarchy and apply aggregation operators to measures
- ○ b. Change the view on the data moving towards the lower levels of a hierarchy and apply aggregation operatore
- ◉ c. Design a hierarchy of dimensions with increasing aggregation level ✖
- ○ d. Design a hierarchy of dimensions with decreasing aggregation level

Your answer is incorrect.

La risposta corretta è:
Change the view on the data moving towards the higher levels of a hierarchy and apply aggregation operators to measures

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about ETL, which of the following activities is related to the **Transformation** step?

Scegli una o più alternative:

- [ ] a. **Snapshot** of the operational data
- [x] b. **Denormalisation** ✓
- [x] c. Calculation of **derived data** ✓
- [ ] d. Usage of dictionaries to solve **inconsistencies**

Risposta corretta.

Le risposte corrette sono: **Denormalisation**, Calculation of **derived data**

Domanda **5**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Link the names of the OLAP operations below to their definitions

Roll-Up	Causes an increase in data aggregation and removes a detail level in a hierarchy ✔
Drill down	Reduces data aggregation and adds a detail level to a hierarchy ✔
Slice and dice	Reduces the number of cube dimensions after setting one of the dimensions to a specific value ✔
Pivot	Changes the layout, in order to analyse a group of data from a different viewpoint ✔
Drill across	Creates a link between concepts in interrelated cubes, in order to compare them ✔

Risposta corretta.

La risposta corretta è: Roll-Up → Causes an increase in data aggregation and removes a detail level in a hierarchy, Drill down → Reduces data aggregation and adds a detail level to a hierarchy, Slice and dice → Reduces the number of cube dimensions after setting one of the dimensions to a specific value, Pivot → Changes the layout, in order to analyse a group of data from a different viewpoint, Drill across → Creates a link between concepts in interrelated cubes, in order to compare them

Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about the general idea of database, what is the purpose of the "Schema on read" strategy?

Scegli una o più alternative:

- [x] a. Possibility to extract data in various shapes ✔
- [ ] b. Optimisation for various types of queries
- [x] c. Flexibility for any kind of query ✔
- [x] d. Avoid preprocessing of data before writing ✔

Risposta corretta.

Le risposte corrette sono: Possibility to extract data in various shapes, Flexibility for any kind of query, Avoid preprocessing of data before writing

Domanda **7**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Check the features of Data Lakes in comparison with Data Warehouses,

Multiple answers allowed. Selecting a wrong option will be penalised

- [x] a. Can run on less expensive HW/SW ✓
- [x] b. Do not enforce data quality ✓
- [ ] c. More complex management
- [ ] d. Completely defined use cases

Your answer is correct.

Le risposte corrette sono:
Can run on less expensive HW/SW,

Do not enforce data quality

Domanda **8**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Stemming and Lemmatization

- (●) a. are text processing techniques to cut suffixes and reduce each word to its root and base form respectively ✓
- ( ) b. are text processing techniques to cut suffixes and reduce each word to its base form and root respectively
- ( ) c. are text processing techniques to cut prefix and reduce each word to its base form and root respectively
- ( ) d. are text processing techniques to cut prefix and reduce each word to its root and base form respectively

Your answer is correct.

La risposta corretta è:
are text processing techniques to cut suffixes and reduce each word to its root and base form respectively

Domanda **9**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Breakeven Point of Precision-Recall is

- a. an interpolated value of a precision-recall graph where precision and recall are identical ✔
- b. an interpolated value of a precision-recall graph where R-precision and recall are identical
- c. a value of a precision-recall graph where precision and recall are identical
- d. a value of a precision-recall graph where R-precision and recall are identical

Your answer is correct.

La risposta corretta è:
an interpolated value of a precision-recall graph where precision and recall are identical

Domanda **10**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In a Chi-square test for feature selection of a term t with respect to a class c, the p-value is 0.015

- a. the term t is not selected as a feature if the confidence level is 0.99 ✔
- b. the term t is selected as a feature if the confidence level is 0.99
- c. the term t is not selected as a feature if the confidence level is 0.95
- d. the term t is selected as a feature if the confidence level is larger than 0.99

Your answer is correct.

La risposta corretta è:
the term t is not selected as a feature if the confidence level is 0.99

Domanda **11**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

a.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

b.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

c.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

d.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

Your answer is correct.

La risposta corretta è:

1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following terms is responsible for the quadratic complexity of the attention mechanism used inside Transformers (i.e., Attention(Q, K, V) = softmax(QK^T / sqrt(d_k))V) ?

a. The product between Q and K^T

b. The product between softmax(QK^T) and V

c. The factor 1/sqrt(d_k)

d. The V matrix

Your answer is correct.

La risposta corretta è:

The product between Q and K^T

Domanda **13**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following is the correct equation of the triplet loss ?

(d(x, y) denotes the euclidean distance between x and y; a, p and n are the embeddings of the anchor, positive and negative sample, respectively)

- a. L = max(0, d(a,p) – d(a,n) + \alpha)
- b. L = max(0, d(a,p) + d(a,n) + \alpha)
- c. L = max(0, \alpha - d(a,p) + d(a,n))
- d. L = max(0, \alpha - d(a,p) - d(a,n))

Your answer is correct.

La risposta corretta è:
L = max(0, d(a,p) – d(a,n) + \alpha)

Domanda **14**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which are the components of a general message-passing layer in a graph neural network?

- a. Message function, Permutation Invariant Aggregator, Differentiable Function
- b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator
- c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator
- d. Message function, Permutation Invariant Aggregator, Attention Mechanism

Your answer is correct.

La risposta corretta è:

Message function, Permutation Invariant Aggregator, Differentiable Function

Domanda **15**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

A "Big Data Loop" consists of

- any iterative procedure which has been designed to be efficient for large datasets which satisfy the definition of Big Data
- a positive feedback loop in which the analysis of data generated by a business allows to make improvements which expands the business, thus generating even more data
- a Big Data generating process which outputs a volume of data such that any analysis program necessarily trails behind, accumulating more and more unprocessed data over time

Your answer is correct.

La risposta corretta è:
a positive feedback loop in which the analysis of data generated by a business allows to make improvements which expands the business, thus generating even more data

Domanda **16**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

High-Performance Computing (HPC) can perform distributed computation. However, it is hard to program an HPC application for big data processing because HPC

- a. requires programmers to control coordination explicitly in programs, using message-passing primitives ✔
- b. places data randomly on the nodes, with random overlaps
- c. is not flexible enough in handling node failures, as it handles them automatically

Your answer is correct.

La risposta corretta è:
requires programmers to control coordination explicitly in programs, using message-passing primitives

◄ Quiz to Data Mining, Text Mining and Big Data Analytics - Theory

Vai a...

Mock Exam ►

Domanda **17**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

In the Hadoop ecosystem, YARN

- is a system which comprises components both for the management of resources and for the management of process containers
- runs a varying number of resource managers in the cluster, depending on the application
- communicates with the application client using a node manager ✖

Your answer is incorrect.

La risposta corretta è:
is a system which comprises components both for the management of resources and for the management of process containers

Domanda **18**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's structured API, if Salary is a column of a data frame Emp, then Salary's values can be displayed by executing

- a.
```
from pyspark.sql.functions import column
Emp.select('Salary').show()
```
- b.
```
from pyspark.sql.functions import column
Emp.columns.select('Salary').show()
```
- c.
```
from pyspark.sql.functions import column
Emp.column('Salary').show()
```

Your answer is correct.

La risposta corretta è:
```
from pyspark.sql.functions import column
Emp.select('Salary').show()
```

Domanda **19**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's structured API, to create a data frame named df from a JSON file named f_name, the following assignment can be used

- a. `df = spark.read.format("json").load(f_name)`
- b. `df = spark.read.load(f_name)`
- c. `df = spark.format("json").load(f_name)`

Your answer is correct.

La risposta corretta è:
```
df = spark.read.format("json").load(f_name)
```

Domanda **20**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The outer join supported by the structured API of Spark

- (selected) may output rows with null values for all the columns of one of the joined tables ✔
- always outputs null values for all the columns of the joined table having the least number of rows
- always outputs rows with null values for all the columns of at least one of the joined tables

Your answer is correct.

La risposta corretta è:
may output rows with null values for all the columns of one of the joined tables

Domanda **21**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's MLlib, transform() is a function that

- (selected) can be used to compute predictions from a machine learning model ✔
- standardises the dataset to have mean 0 and variance 1 in all its numeric columns
- is only used to compute predictions from a machine learning model

Your answer is correct.

La risposta corretta è:
can be used to compute predictions from a machine learning model

Iniziato	Thursday, 19 January 2023, 15:25
Stato	Completato
Terminato	Thursday, 19 January 2023, 16:01
Tempo impiegato	36 min. 15 secondi
Valutazione	**18,50** su un massimo di 21,00 (**88**%)

Domanda **1**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the activities below is part of "Business Understanding" in the CRISP methodology?

a. Which machine learning functions are necessary for my problem?

b. Which data are available?

c. Which data must be collected with a specific campaign?

d. Which are the resources available (manpower, hardware, software, ...) ✔

Your answer is correct.

La risposta corretta è:
Which are the resources available (manpower, hardware, software, ...)

Domanda **2**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following are typical data warehouse queries?

Scegli una o più alternative:

- [x] a. Which products maximize the profit?
- [x] b. What is the total revenue per product category and state?
- [ ] c. What is the total revenue of yesterday in shop "BO03" ?
- [ ] d. Which is the stock of product "XXX" in the "YYY" warehouse
- [x] e. What is the relationship between profits gained by products "WWW" and "ZZZ"?

Risposta corretta.

Le risposte corrette sono: Which products maximize the profit?, What is the total revenue per product category and state?, What is the relationship between profits gained by products "WWW" and "ZZZ"?

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Check the components of the DFM model

Multiple answers allowed. Wrong choices will be penalised

- [x] a. Descriptions of quantitative aspects of facts
- [x] b. Descriptions of aspects of facts allowing a non-numeric, finite domain
- [ ] c. Relationships between facts
- [ ] d. Indexes for fast access

Your answer is correct.

Le risposte corrette sono:
Descriptions of quantitative aspects of facts,

Descriptions of aspects of facts allowing a non-numeric, finite domain

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about ETL, which of the following activities is related to the **Cleansing** step?

Scegli una o più alternative:

- [ ] a. **Snapshot** of the operational data
- [ ] b. Association of a **timestamp** to the operational data
- [x] c. Elimination of **duplicates** ✔
- [x] d. Usage of dictionaries to solve **inconsistencies** ✔

Risposta corretta.

Le risposte corrette sono: Elimination of **duplicates**, Usage of dictionaries to solve **inconsistencies**

Domanda **5**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the definition below describes the OLAP operation **Drill-Down**?

Scegli un'alternativa:

- ( ) a. Creates a link between concepts in interrelated cubes, to compare them
- ( ) b. Causes an increase in data aggregation and removes a detail level in a hierarchy
- (•) c. Reduces data aggregation and adds a detail level to a hierarchy ✔
- ( ) d. Reduces the number of cube dimensions after setting one of the dimensions to a specific value
- ( ) e. Changes the layout, in order to analyse a group of data from a different viewpoint

Risposta corretta.

La risposta corretta è: Reduces data aggregation and adds a detail level to a hierarchy

Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about the general idea of database, what is the meaning of "Schema on write"?

Scegli un'alternativa:

- a. Create a schema for data before writing into the database ✔
- b. Create a schema for data after writing into the database
- c. Change the schema of the data after each write
- d. Create a schema for data while writing into the database

Risposta corretta.

La risposta corretta è: Create a schema for data before writing into the database

Domanda **7**

Parzialmente corretta

Punteggio ottenuto 0,50 su 1,00

Check the features of Data Lakes in comparison with Data Warehouses,

Multiple answers allowed. Selecting a wrong option will be penalised

- [x] a. Can run on less expensive HW/SW ✔
- [x] b. Do not enforce data quality ✔
- [x] c. More complex management ✖
- [ ] d. Completely defined use cases

Your answer is partially correct.

Hai selezionato troppe opzioni.
Le risposte corrette sono:
Can run on less expensive HW/SW,

Do not enforce data quality

Domanda **8**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

According to the definition illustrated, why the TF-IDF of a stop word is zero ?

- a. because the IDF is 1 ✖
- b. because the TF is 1
- c. because the IDF is 0
- d. because the TF is 0

Your answer is incorrect.

La risposta corretta è:
because the IDF is 0

Domanda **9**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The R-precision of a query q measures

- a. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q ✔
- b. the number of relevant documents for the query q in the first R retrieved docs, divided by the number of docs in the corpus
- c. the number of relevant documents for the query q in the first R retrieved docs, divided by R which is the number of docs retrieved
- d. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for each q

Your answer is correct.

La risposta corretta è:
the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q

Domanda **10**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In a Chi-square test for feature selection of a term t with respect to a class c, the p-value is 0.015

- a. the term t is not selected as a feature if the confidence level is 0.99
- b. the term t is selected as a feature if the confidence level is 0.99
- c. the term t is not selected as a feature if the confidence level is 0.95
- d. the term t is selected as a feature if the confidence level is larger than 0.99

Your answer is correct.

La risposta corretta è:
the term t is not selected as a feature if the confidence level is 0.99

Domanda **11**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

a.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

b.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

c.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

d.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

Your answer is correct.

La risposta corretta è:

1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following terms is responsible for the quadratic complexity of the attention mechanism used inside Transformers (i.e., Attention(Q, K, V) = softmax(QK^T / sqrt(d_k))V) ?

- a. The product between Q and K^T ✔
- b. The product between softmax(QK^T) and V
- c. The factor 1/sqrt(d_k)
- d. The V matrix

Your answer is correct.

La risposta corretta è:

The product between Q and K^T

Domanda **13**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

A big e-commerce company wants an information retrieval system to search for product images by providing a text query in natural language. They want their system to be as fast as possible. Which of the following is a sensible approach to address this task ?

a. We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

b. We train a text encoder and an image encoder with metric learning to have a common latent space for text and images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird)

c. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

d. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird)

Your answer is correct.

La risposta corretta è:

We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

Domanda **14**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which are the components of a general message-passing layer in a graph neural network?

a. Message function, Permutation Invariant Aggregator, Differentiable Function ✔

b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator

c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator

d. Message function, Permutation Invariant Aggregator, Attention Mechanism

Your answer is correct.

La risposta corretta è:

Message function, Permutation Invariant Aggregator, Differentiable Function

Domanda **15**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

A "Big Data Loop" consists of

- any iterative procedure which has been designed to be efficient for large datasets which satisfy the definition of Big Data
- a positive feedback loop in which the analysis of data generated by a business allows to make improvements which expands the business, thus generating even more data
- a Big Data generating process which outputs a volume of data such that any analysis program necessarily trails behind, accumulating more and more unprocessed data over time ✖

Your answer is incorrect.

La risposta corretta è:
a positive feedback loop in which the analysis of data generated by a business allows to make improvements which expands the business, thus generating even more data

Domanda **16**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Traditional Relational Data Base Management Systems (RDBMS) can perform distributed computation, but they are unsuitable for big data processing because

- a. their technology is optimised for short accesses to small amounts of data ✔
- b. they are optimised for read-only access to large amounts of data
- c. they cannot operate on GB-sized data efficiently

Your answer is correct.

La risposta corretta è:
their technology is optimised for short accesses to small amounts of data

Domanda **17**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the Hadoop system, YARN

- a. fulfills application client requests to start an application using its resource manager to ask node managers to allocate resources ✔
- b. uses its resource manager to directly create containers
- c. runs one node manager and a varying number of resource managers in the cluster

Your answer is correct.

La risposta corretta è:
fulfills application client requests to start an application using its resource manager to ask node managers to allocate resources

Domanda **18**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The select method of DataFrame, called on a data frame df,

- generates a data frame containing selected rows of df
- generates a data frame containing selected columns of df
- generates a data frame containing selected rows and columns of df

Your answer is correct.

La risposta corretta è:
generates a data frame containing selected columns of df

Domanda **19**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's structured API, DataFrames and Datasets are data collections in the form of tables

- a. for which a schema is defined
- b. with columns of different lengths
- c. with untyped columns

Your answer is correct.

La risposta corretta è:
for which a schema is defined

Domanda **20**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

If df is a Spark DataFrame, df.selectExpr() is a DataFrame method that

- chooses an expression to compute among the expressions passed to the method as arguments
- creates a data frame consisting of every row of df satisfying the expression passed to the method as argument
- creates a data frame such that each of its rows is computed as the values of the sequence of the expressions passed to the method as arguments ✔

Your answer is correct.

La risposta corretta è:
creates a data frame such that each of its rows is computed as the values of the sequence of the expressions passed to the method as arguments

Domanda **21**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's MLlib, transform() is a function that

- can be used to compute predictions from a machine learning model ✔
- standardises the dataset to have mean 0 and variance 1 in all its numeric columns
- is only used to compute predictions from a machine learning model

Your answer is correct.

La risposta corretta è:
can be used to compute predictions from a machine learning model

◄ Quiz to Data Mining, Text Mining and Big Data Analytics - Theory

Vai a...

Mock Exam ►

Started on	Friday, 22 December 2023, 11:32 AM
State	Finished
Completed on	Friday, 22 December 2023, 12:01 PM
Time taken	28 mins 54 secs
Grade	**8.00** out of 16.00 (**50**%)

Question **1**

Incorrect

Mark 0.00 out of 1.00

According to the definition illustrated, why the TF-IDF of a stop word is zero ?

- a. because the IDF is 1 ✖
- b. because the TF is 1
- c. because the IDF is 0
- d. because the TF is 0

Your answer is incorrect.

The correct answer is:
because the IDF is 0

Question **2**

Correct

Mark 1.00 out of 1.00

The R-precision of a query q measures

- a. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q ✔
- b. the number of relevant documents for the query q in the first R retrieved docs, divided by the number of docs in the corpus
- c. the number of relevant documents for the query q in the first R retrieved docs, divided by R which is the number of docs retrieved
- d. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for each q

Your answer is correct.

The correct answer is:
the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q

Question **3**

Incorrect

Mark 0.00 out of 1.00

In a Chi-square test for feature selection of a term t with respect to a class c, the p-value is 0.015

- a. the term t is not selected as a feature if the confidence level is 0.99
- b. the term t is selected as a feature if the confidence level is 0.99 ✖
- c. the term t is not selected as a feature if the confidence level is 0.95
- d. the term t is selected as a feature if the confidence level is larger than 0.99

Your answer is incorrect.

The correct answer is:
the term t is not selected as a feature if the confidence level is 0.99

Question **4**

Incorrect

Mark 0.00 out of 1.00

the latent semantic analysis allows to reduce

- a. the rank of the original term document matrix
- b. the dimensions of the original term document matrix
- c. the rank of the rows of the original term document matrix
- d. the rank of the columns of the original term document matrix

Your answer is incorrect.

The correct answer is:
the rank of the original term document matrix

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following terms is responsible for the quadratic complexity of the attention mechanism used inside Transformers (i.e., Attention(Q, K, V) = softmax(QK^T / sqrt(d_k))V) ?

- a. The product between Q and K^T
- b. The product between softmax(QK^T) and V
- c. The factor 1/sqrt(d_k)
- d. The V matrix

Your answer is correct.

The correct answer is:

The product between Q and K^T

Question **6**

Correct

Mark 1.00 out of 1.00

A big e-commerce company wants an information retrieval system to search for product images by providing a text query in natural language. They want their system to be as fast as possible. Which of the following is a sensible approach to address this task ?

- a. We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)
- b. We train a text encoder and an image encoder with metric learning to have a common latent space for text and images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird)
- c. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)
- d. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird)

Your answer is correct.

The correct answer is:

We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

Question **7**

Incorrect

Mark 0.00 out of 1.00

The Fusion-in-Encoder method is

- a. a merging strategy where the input instance provided to the input layer is concatenated with k external items, for example retrieved by a dense passage retriever, and this single combined input is forwarded to the encoder.

  encoder receives and encodes the input instance, provided to the input layer, that is concatenated with k external items, for example retrieved by a neural layer, and the resulting vecntor is encoded
- b. a merging strategy where the input instance provided to the input layer is combined with k external items, for example retrieved by a dense passage retriever, and these k combined inputs are forwarded to the encoder.
- c. a merging strategy where the input instance provided to the input layer is concatenated with k external items, for example retrieved by a dense passage retriever, and the resulting input is forwarded to the decoder.
- d. a merging strategy where the input instance provided to the input layer is concatenated with k external items, for example retrieved by a dense passage retriever, and these k combined inputs are forwarded to the encoder and decoder. ✖

Your answer is incorrect.

The correct answer is:
a merging strategy where the input instance provided to the input layer is concatenated with k external items, for example retrieved by a dense passage retriever, and this single combined input is forwarded to the encoder.

encoder receives and encodes the input instance, provided to the input layer, that is concatenated with k external items, for example retrieved by a neural layer, and the resulting vecntor is encoded

Question **8**

Incorrect

Mark 0.00 out of 1.00

the dimension of word embedding vectors

- a. is a hyperparameter decided by the data scientist
- b. corresponds to the vocabulary length ✖
- c. corresponds to the number of documents
- d. is automatically defined by the neural network

Your answer is incorrect.

The correct answer is:
is a hyperparameter decided by the data scientist

Question **9**

Correct

Mark 1.00 out of 1.00

The phrase "Big Data Loop" refers to

- a kind of big data processing framework which processes big data using a sequence of phases
- a looping mechanism by which web site data generation accelerates as a consequence of applying analytics to web site data ✔
- an algorithm designed to process big data in an iterative fashion

Your answer is correct.

The correct answer is:
a looping mechanism by which web site data generation accelerates as a consequence of applying analytics to web site data

Question **10**

Correct

Mark 1.00 out of 1.00

High-Performance Computing (HPC) can perform distributed computation. However, it is hard to program an HPC application for big data processing because HPC

- a. requires programmers to control coordination explicitly in programs, using message-passing primitives ✔
- b. places data randomly on the nodes, with random overlaps
- c. is not flexible enough in handling node failures, as it handles them automatically

Your answer is correct.

The correct answer is:
requires programmers to control coordination explicitly in programs, using message-passing primitives

Question **11**

Correct

Mark 1.00 out of 1.00

In the MapReduce programming model

- key-value pairs are the input of the computation and values are the output of the computation
- the map function maps each value to a unique key according to a user supplied mapper
- key-value pairs are transformed into key-value pairs by a user-specified mapper

Your answer is correct.

The correct answer is:
key-value pairs are transformed into key-value pairs by a user-specified mapper

Question **12**

Incorrect

Mark 0.00 out of 1.00

In Spark's structured API, DataFrames and Datasets are data collections in the form of tables

a. with an associated schema
b. with columns of different lengths
c. with untyped columns

Your answer is incorrect.

The correct answer is: with an associated schema

Question **13**

Incorrect

Mark 0.00 out of 1.00

In Spark's structured API, the following assignment creates a data frame named df from a JSON file named f_name:

- a. `df = spark.read.format("json").load(f_name)`
- b. `df = spark.read.load(f_name)`
- c. `df = spark.format("json").load(f_name)`

Your answer is incorrect.

The correct answer is:

`df = spark.read.format("json").load(f_name)`

Question **14**

Correct

Mark 1.00 out of 1.00

In the structured API of Spark, the DataFrame class has a method where(). Executing

```
df.where(arg)
```

where df is DataFrame object, returns a DataFrame object

- with every row of df, except the ones that do not verify expression arg
- with exactly the rows of df which are also in the list passed as arg
- with exactly the rows of df which are identical to the example row passed as arg

Your answer is correct.

The correct answer is:
with every row of df, except the ones that do not verify expression arg

Question **15**

Incorrect

Mark 0.00 out of 1.00

To read a CSV file into a data frame in Spark's Python API

- a. functions format() and load() can be used
- b. the csv.reader() function of Python's csv module must be called ✖
- c. an import csv statement must be used

Your answer is incorrect.

The correct answer is:
functions format() and load() can be used

Question **16**

Correct

Mark 1.00 out of 1.00

In Spark's MLlib, transform() is a function that

- can be used to compute predictions from a machine learning model ✔
- standardises the dataset to have mean 0 and variance 1 in all its numeric columns
- is only used to compute predictions from a machine learning model

Your answer is correct.

The correct answer is:
can be used to compute predictions from a machine learning model

◄ Mock Exam

Jump to...

Quiz to Data Mining, Text Mining and Big Data Analytics - Theory - 2024.01.18 ►

Started on	Thursday, 18 January 2024, 1:31 PM
State	Finished
Completed on	Thursday, 18 January 2024, 2:00 PM
Time taken	29 mins 1 sec
Grade	**12.00** out of 16.00 (**75**%)

Question **1**

Correct

Mark 1.00 out of 1.00

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

a.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

b.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

c.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

d.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

Your answer is correct.

The correct answer is:
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

Question **2**

Not answered

Marked out of 1.00

In the MapReduce programming model

- key-value pairs are the input of the computation and values are the output of the computation
- the map function maps each value to a unique key according to a user supplied mapper
- key-value pairs are transformed into key-value pairs by a user-specified mapper

Your answer is incorrect.

The correct answer is:
key-value pairs are transformed into key-value pairs by a user-specified mapper

Question **3**

Correct

Mark 1.00 out of 1.00

Traditional Relational Data Base Management Systems (RDBMS) can perform distributed computation, but they are unsuitable for big data processing because

- a. their technology is optimised for short accesses to small amounts of data ✔
- b. they are optimised for read-only access to large amounts of data
- c. they cannot operate on GB-sized data efficiently

Your answer is correct.

The correct answer is:
their technology is optimised for short accesses to small amounts of data

Question **4**

Correct

Mark 1.00 out of 1.00

In Spark's structured API, DataFrames and Datasets are data collections in the form of tables

- a. with an associated schema
- b. with columns of different lengths
- c. with untyped columns

Your answer is correct.

The correct answer is: with an associated schema

Question **5**

Incorrect

Mark 0.00 out of 1.00

Breakeven Point of Precision-Recall is

- a. an interpolated value of a precision-recall graph where precision and recall are identical
- b. an interpolated value of a precision-recall graph where R-precision and recall are identical
- c. a value of a precision-recall graph where precision and recall are identical
- d. a value of a precision-recall graph where R-precision and recall are identical

Your answer is incorrect.

The correct answer is:
an interpolated value of a precision-recall graph where precision and recall are identical

Question **6**

Correct

Mark 1.00 out of 1.00

The Fusion-in-Encoder method is

- a. a merging strategy where the input instance provided to the input layer is concatenated with k external items, for example retrieved by a dense passage retriever, and this single combined input is forwarded to the encoder.

  encoder receives and encodes the input instance, provided to the input layer, that is concatenated with k external items, for example retrieved by a neural layer, and the resulting vecntor is encoded
- b. a merging strategy where the input instance provided to the input layer is combined with k external items, for example retrieved by a dense passage retriever, and these k combined inputs are forwarded to the encoder.
- c. a merging strategy where the input instance provided to the input layer is concatenated with k external items, for example retrieved by a dense passage retriever, and the resulting input is forwarded to the decoder.
- d. a merging strategy where the input instance provided to the input layer is concatenated with k external items, for example retrieved by a dense passage retriever, and these k combined inputs are forwarded to the encoder and decoder.

Your answer is correct.

The correct answer is:
a merging strategy where the input instance provided to the input layer is concatenated with k external items, for example retrieved by a dense passage retriever, and this single combined input is forwarded to the encoder.

encoder receives and encodes the input instance, provided to the input layer, that is concatenated with k external items, for example retrieved by a neural layer, and the resulting vecntor is encoded

Question **7**

Correct

Mark 1.00 out of 1.00

The function to create a decision tree classifier object in MLlib is

- a. `DecisionTreeClassifier()`
- b. `DecisionTreeClassifier().transform()`
- c. `DecisionTreeClassifier().fit()`

Your answer is correct.

The correct answer is:
`DecisionTreeClassifier()`

Question **8**

Incorrect

Mark 0.00 out of 1.00

Which of the following is the correct contrastive loss ?

(d(v, w) denotes the euclidean distance between v and w; x1, x2 are the embeddings of two inputs and y1 and y2 their two classes respectively, \alpha is the margin)

- a. L = d(x1, x2) if y1 == y2
  L = max(0, \alpha - d(x1, x2)) if y1 ≠ y2
- b. L = d(x1, x2) if y1 ≠ y2
  L = max(0, \alpha - d(x1, x2)) if y1 == y2
- c. L = d(x1, x2) if y1 ≠ y2
  L = max(0, \alpha + d(x1, x2)) if y1 == y2 ✖
- d. L = d(x1, x2) if y1 == y2
  L = min(0, d(x1, x2) - \alpha) if y1 ≠ y2

Your answer is incorrect.

The correct answer is:
L = d(x1, x2) if y1 == y2
L = max(0, \alpha - d(x1, x2)) if y1 ≠ y2

Question **9**

Correct

Mark 1.00 out of 1.00

In Spark's MLlib, transform() is a function that

- can be used to compute predictions from a machine learning model ✔
- standardises the dataset to have mean 0 and variance 1 in all its numeric columns
- is only used to compute predictions from a machine learning model

Your answer is correct.

The correct answer is:
can be used to compute predictions from a machine learning model

Question **10**

Correct

Mark 1.00 out of 1.00

In the Hadoop ecosystem, YARN

- (selected) is a system which comprises components both for the management of resources and for the management of process containers ✔
- runs a varying number of resource managers in the cluster, depending on the application
- communicates with the application client using a node manager

Your answer is correct.

The correct answer is:
is a system which comprises components both for the management of resources and for the management of process containers

Question **11**

Incorrect

Mark 0.00 out of 1.00

In a Chi-square test for feature selection of a term t with respect to a class c, the p-value is 0.015

- a. the term t is selected as a feature if the confidence level is 0.98
- b. the term t is selected as a feature if the confidence level is 0.99
- c. (selected) the term t is not selected as a feature if the confidence level is 0.95 ✘
- d. the term t is selected as a feature if the confidence level is larger than 0.99

Your answer is incorrect.

The correct answer is:
the term t is selected as a feature if the confidence level is 0.98

Question **12**

Correct

Mark 1.00 out of 1.00

The phrase "Big Data Loop" refers to

- a kind of big data processing framework which processes big data using a sequence of phases
- a looping mechanism by which web site data generation accelerates as a consequence of applying analytics to web site data ✔
- an algorithm designed to process big data in an iterative fashion

Your answer is correct.

The correct answer is:
a looping mechanism by which web site data generation accelerates as a consequence of applying analytics to web site data

Question **13**

Correct

Mark 1.00 out of 1.00

In the structured API of Spark, the DataFrame class has a method where(). Executing

```
df.where(arg)
```

where df is DataFrame object, returns a DataFrame object

- with every row of df, except the ones that do not verify expression arg ✔
- with exactly the rows of df which are also in the list passed as arg
- with exactly the rows of df which are identical to the example row passed as arg

Your answer is correct.

The correct answer is:
with every row of df, except the ones that do not verify expression arg

Question **14**

Correct

Mark 1.00 out of 1.00

Which are the components of a general message-passing layer in a graph neural network?

- a. Message function, Permutation Invariant Aggregator, Differentiable Function ✔
- b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator
- c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator
- d. Message function, Permutation Invariant Aggregator, Attention Mechanism

Your answer is correct.

The correct answer is:

Message function, Permutation Invariant Aggregator, Differentiable Function

Question **15**

Correct

Mark 1.00 out of 1.00

According to the definition illustrated, why the TF-IDF of a stop word is zero ?

- a. because the IDF is 1
- b. because the TF is 1
- c. because the IDF is 0 ✔
- d. because the TF is 0

Your answer is correct.

The correct answer is:
because the IDF is 0

Question **16**

Correct

Mark 1.00 out of 1.00

Transformers have been introduced for natural language processing tasks. Which is the correct sequence of layers in a typical Transformer Encoder (without considering residual connections and normalization layers) ?

- a. Input embeddings -> Position Embeddings -> MultiHead Attention -> Feed Forward
- b. Feed Forward -> Input embeddings -> Position Embeddings -> MultiHead Attention
- c. Position Embeddings -> Input embeddings -> MultiHead Attention -> Feed Forward
- d. Input embeddings -> Position Embeddings -> Feed Forward -> MultiHead Attention

Your answer is correct.

The correct answer is:

Input embeddings -> Position Embeddings -> MultiHead Attention -> Feed Forward

◄ Quiz to Data Mining, Text Mining and Big Data Analytics - Theory - 2023.12.22

Jump to...

Iniziato	Wednesday, 21 December 2022, 12:18
Stato	Completato
Terminato	Wednesday, 21 December 2022, 13:00
Tempo impiegato	41 min. 59 secondi
Valutazione	Non ancora valutato

Domanda **1**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the activities below is part of "Business Understanding" in the CRISP methodology?

- a. Which machine learning functions are necessary for my problem?
- b. Which data are available?
- c. Which data must be collected with a specific campaign?
- d. Which are the resources available (manpower, hardware, software, ...) ✔

Your answer is correct.

La risposta corretta è:
Which are the resources available (manpower, hardware, software, ...)

Domanda **2**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Select the sentences describing a characteristic of a *Data Warehouse*

Scegli una o più alternative:

- [x] a. Non volatile ✔
- [ ] b. Constantly updated as soon as the base data are updated
- [ ] c. Includes all the base data in their native format
- [x] d. Includes the time dimension ✔
- [x] e. Solves the inconsistencies ✔

Risposta corretta.

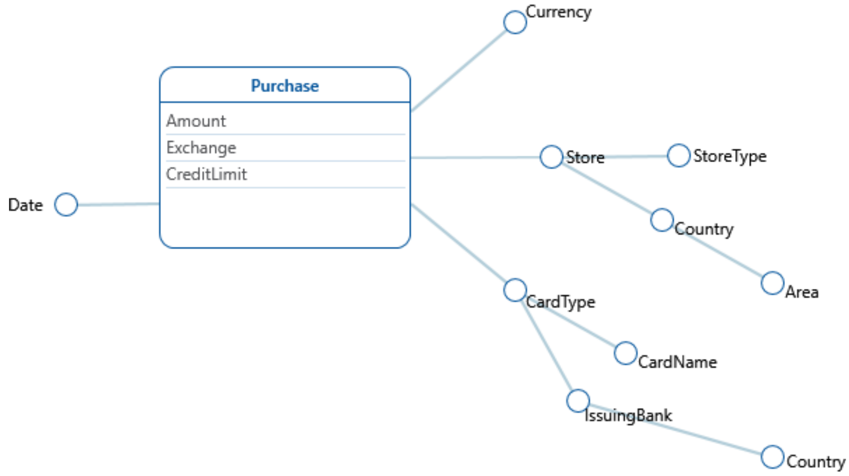
Le risposte corrette sono: Non volatile, Includes the time dimension, Solves the inconsistencies

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the texts describes the DFM schema below?



Scegli un'alternativa:

- a. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country. ✔
- b. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores types have also a country and an area. Purchases have a card type (e.g. credit or debit) and a card name (e.g. MyCard or JeansCard). Card names have also an issuing bank and a country.
- c. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type and country; stores have also an area. Purchases have a card name (e.g. MyCard or JeansCard), card_names have a card type (e.g. credit or debit). Card types have also an issuing bank and a country.

Risposta corretta.

La risposta corretta è: We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country.

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about ETL, which of the following activities is related to the **Extraction** step?

Scegli una o più alternative:

- [x] a. **Snapshot** of the operational data ✓
- [x] b. Association of a **timestamp** to the operational data ✓
- [ ] c. Elimination of **duplicates**
- [ ] d. Usage of dictionaries to solve **inconsistencies**

Risposta corretta.

Le risposte corrette sono: **Snapshot** of the operational data, Association of a **timestamp** to the operational data

Domanda **5**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the definition below describes the OLAP operation **Drill-Down**?

Scegli un'alternativa:

- ○ a. Creates a link between concepts in interrelated cubes, to compare them
- ○ b. Causes an increase in data aggregation and removes a detail level in a hierarchy
- ◉ c. Reduces data aggregation and adds a detail level to a hierarchy ✓
- ○ d. Reduces the number of cube dimensions after setting one of the dimensions to a specific value
- ○ e. Changes the layout, in order to analyse a group of data from a different viewpoint

Risposta corretta.

La risposta corretta è: Reduces data aggregation and adds a detail level to a hierarchy

Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about the general idea of database, what is the meaning of "Schema on read"?

Scegli un'alternativa:

- a. Write the raw data in the database, then at reading time shape then according to the reader's needs ✔
- b. Create a schema for data after reading them from the database
- c. Change the schema of the data before each read
- d. Read the schema of the data before each data read

Risposta corretta.

Le risposte corrette sono: Write the raw data in the database, then at reading time shape then according to the reader's needs, Create a schema for data after reading them from the database

Domanda **7**

Completo

Punteggio max.: 1,00

Describe the main principles and characteristics of a Data Warehouse (not more than 300 words)

Data Warehouse (DWH) is an optimized central repository that stores data from several data sources. It is a core component of BI, since it allows to perform data analyses and generating report by exploiting its multidimensional model. A DWH is a decision support system. indeed it helps in decision-making. For this reason, we do not have the same level of detail we have in transactional databases, but we have an aggregated view across discrete dimensions.

It is modeled in a subject-oriented way: the main focus is on the enterprise concepts

Data from several sources can be heterogeneous (both structured and unstructured).

Changes in the source data are tracked and recorded in an incremental way (updates and deletes do not modify data in the DWH). Therefore, it add a time dimension to data, allowing to perform historical analyses.

In a DWH, we want scalability (hw and sw can be easily scaled on needs), extensibility (we add new concept with low effort). security and administrability.

Domanda **8**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

Why the TF-IDF of a stop word is zero ?

- a. because the IDF is 1
- b. because the TF is 1
- c. because the IDF is 0 ✖
- d. because the TF is 0

Your answer is incorrect.

La risposta corretta è:
because the IDF is 1

Domanda **9**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The R-precision of a query q measures

- a. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q ✔
- b. the number of relevant documents for the query q in the first R retrieved docs, divided by the number of docs in the corpus
- c. the number of relevant documents for the query q in the first R retrieved docs, divided by R which is the number of docs retrieved
- d. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for each q

Your answer is correct.

La risposta corretta è:
the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q

Domanda **10**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In a Chi-square test for feature selection of a term t with respect to a class c, the p-value is 0.015

- a. the term t is not selected as a feature if the confidence level is 0.99
- b. the term t is selected as a feature if the confidence level is 0.99
- c. the term t is not selected as a feature if the confidence level is 0.95
- d. the term t is selected as a feature if the confidence level is larger than 0.99

Your answer is correct.

La risposta corretta è:
the term t is not selected as a feature if the confidence level is 0.99

Domanda **11**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

a.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

b.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

c.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

d.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

Your answer is correct.

La risposta corretta è:

1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following terms is responsible for the quadratic complexity of the attention mechanism used inside Transformers (i.e., Attention(Q, K, V) = softmax(QK^T / sqrt(d_k))V) ?

- a. The product between Q and K^T ✔
- b. The product between softmax(QK^T) and V
- c. The factor 1/sqrt(d_k)
- d. The V matrix

Your answer is correct.

La risposta corretta è:

The product between Q and K^T

Domanda **13**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following is the correct equation of the triplet loss ?

(d(x, y) denotes the euclidean distance between x and y; a, p and n are the embeddings of the anchor, positive and negative sample, respectively)

- a. L = max(0, d(a,p) – d(a,n) + \alpha) ✔
- b. L = max(0, d(a,p) + d(a,n) + \alpha)
- c. L = max(0, \alpha - d(a,p) + d(a,n))
- d. L = max(0, \alpha - d(a,p) - d(a,n))

Your answer is correct.

La risposta corretta è:
L = max(0, d(a,p) – d(a,n) + \alpha)

Domanda **14**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which are the components of a general message-passing layer in a graph neural network?

- a. Message function, Permutation Invariant Aggregator, Differentiable Function
- b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator
- c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator
- d. Message function, Permutation Invariant Aggregator, Attention Mechanism

Your answer is correct.

La risposta corretta è:

Message function, Permutation Invariant Aggregator, Differentiable Function

Domanda 15

Risposta corretta

Quiz to Data Mining, Text Mining and Big Data Analytics - Theory [copy] ►

Punteggio ottenuto 1,00 su 1,00

The phrase "Big Data Loop" refers to

- a kind of big data processing framework which processes big data using a sequence of phases
- the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites
- an algorithm designed to process big data in an iterative fashion

Your answer is correct.

La risposta corretta è:
the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites

Domanda **16**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The filter() method of DataFrame in Spark's structured API

- retrieves specified columns of a data frame for its rows which satisfy a boolean condition
- selects specified, possibly renamed, columns of a data frame
- accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition ✔

Your answer is correct.

La risposta corretta è:
accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition

Domanda **17**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's structured API, calling the join method on a data frame df, with df1 as argument

- creates a new data frame containing all columns of df and df1; the computation is legal only if df and df1 have the same number of rows
- creates a new data frame containing all rows of df and df1; the computation is legal only if df and df1 have the same number of columns
- creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df ✔

Your answer is correct.

La risposta corretta è:
creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df

Domanda **18**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The map function in MapReduce

- ○ always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs
- ○ is a predefined function which processes key-value pairs; its behaviour can be changed by passing parameters
- ◉ is a user defined function which is applied to every key-value pair of the dataset ✔

Your answer is correct.

Le risposte corrette sono:
always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs,

is a user defined function which is applied to every key-value pair of the dataset

Domanda **19**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the Hadoop system, the MapReduce application client

- ○ centrally computes a map function, then negotiates resource allocation with YARN for the Reduce phase
- ○ autonomously computes the resources each worker is allowed to use
- ◉ sends requests to YARN for resource allocation ✔

Your answer is correct.

La risposta corretta è:
sends requests to YARN for resource allocation

Domanda **20**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

A difference between an RDBMS and MapReduce is

- that an RDBMS is capable of performing distributed computation
- that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses ✔
- that MapReduce cannot compute any form of aggregation query on the data

Your answer is correct.

La risposta corretta è:
that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses

Domanda **21**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the structured API of Spark, pyspark.sql.Row

- is method of DataFrameReader which gets one row at a time from a JSON text file
- can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame ✔
- is method for reading rows from a data frame

Your answer is correct.

La risposta corretta è:
can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame

Iniziato	Wednesday, 21 December 2022, 12:18
Stato	Completato
Terminato	Wednesday, 21 December 2022, 12:59
Tempo impiegato	40 min. 49 secondi
Valutazione	Non ancora valutato

manda **1**
posta errata
nteggio
enuto 0,00 su
0

In which part of the CRISP methodology we perform the **test design** activity?

- a. Modeling
- b. Evaluation ✖
- c. Business Understanding
- d. Data Understanding

Your answer is incorrect.

La risposta corretta è:
Modeling

manda **2**
posta
retta
nteggio
enuto 1,00 su
0

Which of the following sentences describes an advantage of a Data Warehouse with respect to a standard DBMS

Scegli una o più alternative:

- a. Allows analysis along the time dimension ✔
- b. Allows efficient execution of key-based queries
- c. Allows efficient execution of multi-dimensional queries ✔
- d. Has tools for helping to solve inconsistencies ✔
- e. Manages efficiently data updates

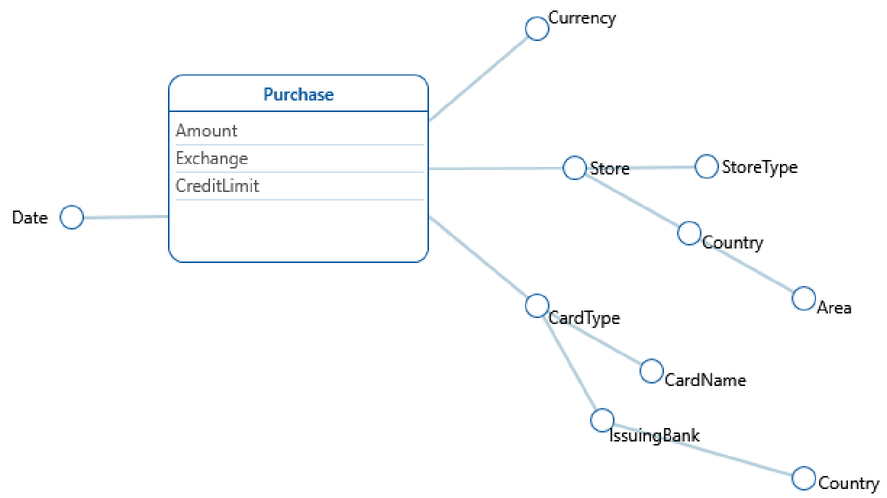
Risposta corretta.

Le risposte corrette sono: Allows analysis along the time dimension, Allows efficient execution of multi-dimensional queries, Has tools for helping to solve inconsistencies

manda **3**
posta

Which of the texts describes the DFM schema below?

Which of the texts describes the DFM schema below?



retta

nteggio
enuto 1,00 su
0

Scegli un'alternativa:

- a. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country.
- b. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores types have also a country and an area. Purchases have a card type (e.g. credit or debit) and a card name (e.g. MyCard or JeansCard). Card names have also an issuing bank and a country.
- c. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type and country; stores have also an area. Purchases have a card name (e.g. MyCard or JeansCard), card_names have a card type (e.g. credit or debit). Card types have also an issuing bank and a country.

Risposta corretta.

La risposta corretta è: We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country.

manda **4**
posta
retta

nteggio
enuto 1,00 su
0

Talking about ETL, which of the following activities is related to the **Cleansing** step?

Scegli una o più alternative:

- a. **Snapshot** of the operational data
- b. Association of a **timestamp** to the operational data
- c. Elimination of **duplicates**
- d. Usage of dictionaries to solve **inconsistencies**

Risposta corretta.

Le risposte corrette sono: Elimination of **duplicates**, Usage of dictionaries to solve **inconsistencies**

manda 5
posta
retta
nteggio
enuto 1,00 su
0

Link the names of the OLAP operations below to their definitions

Roll-Up	Causes an increase in data aggregation and removes a detail level in a hierarchy
Drill down	Reduces data aggregation and adds a detail level to a hierarchy
Slice and dice	Reduces the number of cube dimensions after setting one of the dimensions to a specific value
Pivot	Changes the layout, in order to analyse a group of data from a different viewpoint
Drill across	Creates a link between concepts in interrelated cubes, in order to compare them

Risposta corretta.

La risposta corretta è: Roll-Up → Causes an increase in data aggregation and removes a detail level in a hierarchy, Drill down → Reduces data aggregation and adds a detail level to a hierarchy, Slice and dice → Reduces the number of cube dimensions after setting one of the dimensions to a specific value, Pivot → Changes the layout, in order to analyse a group of data from a different viewpoint, Drill across → Creates a link between concepts in interrelated cubes, in order to compare them

manda 6
posta errata
nteggio
enuto 0,00 su
0

Talking about the general idea of database, what is the meaning of "Schema on write"?

Scegli un'alternativa:

- a. Create a schema for data before writing into the database
- b. Create a schema for data after writing into the database ✖
- c. Change the schema of the data after each write
- d. Create a schema for data while writing into the database

Risposta errata.

La risposta corretta è: Create a schema for data before writing into the database

manda 7
mpleto
nteggio max :

Describe the main objectives and benefits of the CRISP

nteggio max.. 0

methodology

The main objectives of the CRISP methodology are

- the business understanding,
- the data understanding,
- data transform
- model design
- evaluation
- deploy

manda **8**

posta retta

nteggio enuto 1,00 su 0

Stemming and Lemmatization

a. are text processing techniques to cut suffixes and reduce each word to its root and base form respectively ✔

b. are text processing techniques to cut suffixes and reduce each word to its base form and root respectively

c. are text processing techniques to cut prefix and reduce each word to its base form and root respectively

d. are text processing techniques to cut prefix and reduce each word to its root and base form respectively

Your answer is correct.

La risposta corretta è:
are text processing techniques to cut suffixes and reduce each word to its root and base form respectively

manda **9**

posta errata

nteggio enuto 0,00 su 0

Breakeven Point of Precision-Recall is

a. an interpolated value of a precision-recall graph where precision and recall are identical

b. an interpolated value of a precision-recall graph where R-precision and recall are identical ✖

c. a value of a precision-recall graph where precision and recall are identical

d. a value of a precision-recall graph where R-precision and recall are identical

Your answer is incorrect.

La risposta corretta è:

an interpolated value of a precision-recall graph where precision and recall are identical

manda **10**

posta errata

nteggio
enuto 0,00 su
0

In a Chi-square test for feature selection of a term t with respect to a class c, the p-value is 0.015

- a. the term t is not selected as a feature if the confidence level is 0.99
- b. the term t is selected as a feature if the confidence level is 0.99 ✖
- c. the term t is not selected as a feature if the confidence level is 0.95
- d. the term t is selected as a feature if the confidence level is larger than 0.99

Your answer is incorrect.

La risposta corretta è:
the term t is not selected as a feature if the confidence level is 0.99

manda **11**

posta errata

nteggio
enuto 0,00 su
0

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

a.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

b. ✖
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

c.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q

2. term weighting of the onehot encoded vector q

3. fold-in of the term weighted vector q in the LSA space

4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues

5. return the top-k highest cosine similarity LSA vector documents

d.
1. onehot encoding vector of q with respect to the dictionary

2. term weighting of the onehot encoded vector q

3. fold-in of the term weighted vector q in the LSA space

4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues

5. return the top-k highest cosine similarity LSA vector documents

Your answer is incorrect.

La risposta corretta è:
1. onehot encoding vector of q with respect to the dictionary

2. term weighting of the onehot encoded vector q

3. fold-in of the term weighted vector q in the LSA space

4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues

5. return the top-k highest cosine similarity LSA vector terms

manda **12**

posta
retta

nteggio
enuto 1,00 su
0

Which of the following terms is responsible for the quadratic complexity of the attention mechanism used inside Transformers (i.e., Attention(Q, K, V) = softmax(QK^T / sqrt(d_k))V) ?

a. The product between Q and K^T ✔

b. The product between softmax(QK^T) and V

c. The factor 1/sqrt(d_k)

d. The V matrix

Your answer is correct.

La risposta corretta è:

The product between Q and K^T

manda **13**
posta errata
nteggio
enuto 0,00 su
0

Which of the following is the correct equation of the triplet loss ?

(d(x, y) denotes the euclidean distance between x and y; a, p and n are the embeddings of the anchor, positive and negative sample, respectively)

- a. L = max(0, d(a,p) – d(a,n) + \alpha)
- b. L = max(0, d(a,p) + d(a,n) + \alpha)
- c. L = max(0, \alpha - d(a,p) + d(a,n)) ✖
- d. L = max(0, \alpha - d(a,p) - d(a,n))

Your answer is incorrect.

La risposta corretta è:
L = max(0, d(a,p) – d(a,n) + \alpha)

manda **14**
posta errata
nteggio
enuto 0,00 su
0

Which are the components of a general message-passing layer in a graph neural network?

- a. Message function, Permutation Invariant Aggregator, Differentiable Function
- b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator
- c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator ✖
- d. Message function, Permutation Invariant Aggregator, Attention Mechanism

Your answer is incorrect.

La risposta corretta è:

Message function, Permutation Invariant Aggregator, Differentiable Function

manda **15**
posta
retta
nteggio
enuto 1,00 su
0

The phrase "Big Data Loop" refers to

- ○ a kind of big data processing framework which processes big data using a sequence of phases
- ◉ the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites ✔
- ○ an algorithm designed to process big data in an iterative fashion

Your answer is correct.

La risposta corretta è:
the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites

manda **16**
posta errata
nteggio
enuto 0,00 su
0

The filter() method of DataFrame in Spark's structured API

- ◉ retrieves specified columns of a data frame for its rows which satisfy a boolean condition ✖
- ○ selects specified, possibly renamed, columns of a data frame
- ○ accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition

Your answer is incorrect.

La risposta corretta è:
accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition

manda **17**
posta
retta
nteggio
enuto 1,00 su
0

In Spark's structured API, calling the join method on a data frame df, with df1 as argument

- ○ creates a new data frame containing all columns of df and df1; the computation is legal only if df and df1 have the same number of rows
- ○ creates a new data frame containing all rows of df and df1; the computation is legal only if df and df1 have the same number of columns
- ◉ creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df ✔

Your answer is correct.

La risposta corretta è:
creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df

manda **18**
posta
retta
nteggio
enuto 1,00 su
0

The map function in MapReduce

- always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs
- is a predefined function which processes key-value pairs; its behaviour can be changed by passing parameters
- is a user defined function which is applied to every key-value pair of the dataset ✔

Your answer is correct.

Le risposte corrette sono:
always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs,

is a user defined function which is applied to every key-value pair of the dataset

manda **19**
posta
retta
nteggio
enuto 1,00 su
0

In the Hadoop system, the MapReduce application client

- centrally computes a map function, then negotiates resource allocation with YARN for the Reduce phase
- autonomously computes the resources each worker is allowed to use
- sends requests to YARN for resource allocation ✔

Your answer is correct.

La risposta corretta è:
sends requests to YARN for resource allocation

manda **20**
posta
retta
nteggio
enuto 1,00 su
0

A difference between an RDBMS and MapReduce is

- that an RDBMS is capable of performing distributed computation
- that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses ✔
- that MapReduce cannot compute any form of aggregation query on the data

Your answer is correct.

La risposta corretta è:
that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses

manda **21**

posta
retta

nteggio
enuto 1,00 su
0

In the structured API of Spark, pyspark.sql.Row

- is method of DataFrameReader which gets one row at a time from a JSON text file
- can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame ✔
- is method for reading rows from a data frame

Your answer is correct.

La risposta corretta è:
can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame

Iniziato	Wednesday, 21 December 2022, 12:17
Stato	Completato
Terminato	Wednesday, 21 December 2022, 12:54
Tempo impiegato	36 min. 44 secondi
Valutazione	Non ancora valutato

Domanda **1**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Which of the activities below is part of "Business Understanding" in the CRISP methodology?

a. Which machine learning functions are necessary for my problem?
b. Which data are available?
c. Which data must be collected with a specific campaign?
d. Which are the resources available (manpower, hardware, software, ...)

Your answer is correct.

La risposta corretta è:
Which are the resources available (manpower, hardware, software, ...)

Domanda **2**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Associate the various Data Warehouse Architectures to the respective advantages

Single Layer: The occupation of space is minimised

Two Layers: The workloads for analysis and daily operations are separ

Three Layers: An intermediate level of consolidated data is available

Risposta corretta.

La risposta corretta è:
Associate the various Data Warehouse Architectures to the respective advantages

Single Layer: [The occupation of space is minimised]

Two Layers: [The workloads for analysis and daily operations are separated]

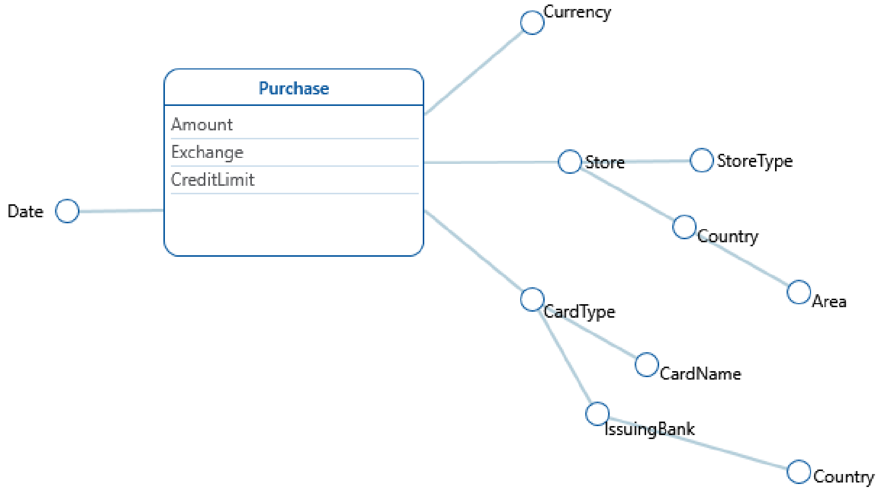
Three Layers: [An intermediate level of consolidated data is available]

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the texts describes the DFM schema below?



Scegli un'alternativa:

- a. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country. ✔
- b. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores types have also a country and an area. Purchases have a card type (e.g. credit or debit) and a card name (e.g. MyCard or JeansCard). Card names have also an issuing bank and a country.
- c. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type and country; stores have also an area. Purchases have a card name (e.g. MyCard or JeansCard), card_names have a card type (e.g. credit or debit). Card types have also an issuing bank and a country.

Risposta corretta.

La risposta corretta è: We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country.

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about ETL, which of the following activities is related to the **Transformation** step?

Scegli una o più alternative:

- [ ] a. **Snapshot** of the operational data
- [x] b. **Denormalisation** ✔
- [x] c. Calculation of **derived data** ✔
- [ ] d. Usage of dictionaries to solve **inconsistencies**

Risposta corretta.

Le risposte corrette sono: **Denormalisation**, Calculation of **derived data**

Domanda **5**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the definition below describes the OLAP operation **Drill-Down**?

Scegli un'alternativa:

- a. Creates a link between concepts in interrelated cubes, to compare them
- b. Causes an increase in data aggregation and removes a detail level in a hierarchy
- c. Reduces data aggregation and adds a detail level to a hierarchy ✔
- d. Reduces the number of cube dimensions after setting one of the dimensions to a specific value
- e. Changes the layout, in order to analyse a group of data from a different viewpoint

Risposta corretta.

La risposta corretta è: Reduces data aggregation and adds a detail level to a hierarchy

Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about the general idea of database, what is the meaning of "Schema on write"?

Scegli un'alternativa:

- a. Create a schema for data before writing into the database ✔
- b. Create a schema for data after writing into the database
- c. Change the schema of the data after each write
- d. Create a schema for data while writing into the database

Risposta corretta.

La risposta corretta è: Create a schema for data before writing into the database

Domanda **7**

Completo

Punteggio max.: 1,00

Describe the main principles and characteristics of a Data Warehouse (not more than 300 words)

Data Warehouses (DW) are a type of Decision Support System. In its most basic form it is a snapshot of a buiseness operational data specifically designed to support analyses and queries. DW has the following features:

-Subject oriented: they are focused on companies specific entities such as products, sale etc.

-Uniform: they integrate data from heterogeneous sources presenting an unified view

-Non-volatile: Data in datawarehouses are never deleted or updated

Avoiding the deletion of data, DW enables an historical view of the data.

Their tipical workflow consist on a few read only queries but they involve a huge amount of data.

DW should also be scalable and flexible, secure and easy to manage.

There are different types of architecture for datawarehouses:

-Single Layer: the occupation of space is minimized and the DW is implemented trough a middleware,

-Two Layers: separetes the loading and operation tasks

-Three Layers: add a layer called reconciled layer which is an intermediate layer for consolitdated data.

Domanda **8**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

Why the TF-IDF of a stop word is zero ?

- a. because the IDF is 1
- b. because the TF is 1
- c. because the IDF is 0 ✖
- d. because the TF is 0

Your answer is incorrect.

La risposta corretta è:
because the IDF is 1

Domanda **9**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Breakeven Point of Precision-Recall is

- a. an interpolated value of a precision-recall graph where precision and recall are identical ✔
- b. an interpolated value of a precision-recall graph where R-precision and recall are identical
- c. a value of a precision-recall graph where precision and recall are identical
- d. a value of a precision-recall graph where R-precision and recall are identical

Your answer is correct.

La risposta corretta è:
an interpolated value of a precision-recall graph where precision and recall are identical

Domanda **10**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

In a Chi-square test for feature selection of a term t with respect to a class c, the p-value is 0.015

a. the term t is not selected as a feature if the confidence level is 0.99

b. the term t is selected as a feature if the confidence level is 0.99 ✖

c. the term t is not selected as a feature if the confidence level is 0.95

d. the term t is selected as a feature if the confidence level is larger than 0.99

Your answer is incorrect.

La risposta corretta è:
the term t is not selected as a feature if the confidence level is 0.99

Domanda **11**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

a. 1. onehot encoding vector of q with respect to the dictionary ✔

2. term weighting of the onehot encoded vector q

3. fold-in of the term weighted vector q in the LSA space

4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues

5. return the top-k highest cosine similarity LSA vector terms

b. 1. onehot encoding vector of q with respect to the dictionary

2. term weighting of the onehot encoded vector q

3. fold-in of the term weighted vector q in the LSA space

4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues

5. return the top-k highest cosine similarity LSA vector terms

c. 1. onehot encoding vector of q with respect to the dictionary

2. term weighting of the onehot encoded vector q

3. fold-in of the term weighted vector q in the LSA space

4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues

5. return the top-k highest cosine similarity LSA vector documents

d. 1. onehot encoding vector of q with respect to the dictionary

2. term weighting of the onehot encoded vector q

3. fold-in of the term weighted vector q in the LSA space

4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues

5. return the top-k highest cosine similarity LSA vector documents

Your answer is correct.

La risposta corretta è:
1. onehot encoding vector of q with respect to the dictionary

2. term weighting of the onehot encoded vector q

3. fold-in of the term weighted vector q in the LSA space

4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues

5. return the top-k highest cosine similarity LSA vector terms

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following terms is responsible for the quadratic complexity of the attention mechanism used inside Transformers (i.e., Attention(Q, K, V) = softmax(QK^T / sqrt(d_k))V) ?

a. The product between Q and K^T

b. The product between softmax(QK^T) and V

c. The factor 1/sqrt(d_k)

d. The V matrix

Your answer is correct.

La risposta corretta è:

The product between Q and K^T

Domanda **13**

Risposta corretta

Punteggio

Which of the following is the correct equation of the triplet loss ?

ottenuto 1,00 su 1,00

(d(x, y) denotes the euclidean distance between x and y; a, p and n are the embeddings of the anchor, positive and negative sample, respectively)

- a. L = max(0, d(a,p) – d(a,n) + \alpha) ✔
- b. L = max(0, d(a,p) + d(a,n) + \alpha)
- c. L = max(0, \alpha - d(a,p) + d(a,n))
- d. L = max(0, \alpha - d(a,p) - d(a,n))

Your answer is correct.

La risposta corretta è:
L = max(0, d(a,p) – d(a,n) + \alpha)

Domanda **14**
Risposta errata
Punteggio ottenuto 0,00 su 1,00

Which are the components of a general message-passing layer in a graph neural network?

- a. Message function, Permutation Invariant Aggregator, Differentiable Function
- b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator
- c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator ✖
- d. Message function, Permutation Invariant Aggregator, Attention Mechanism

Your answer is incorrect.

La risposta corretta è:

Message function, Permutation Invariant Aggregator, Differentiable Function

Domanda **15**
Risposta

The phrase "Big Data Loop" refers to

corretta

Punteggio ottenuto 1,00 su 1,00

- a kind of big data processing framework which processes big data using a sequence of phases
- the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites
- an algorithm designed to process big data in an iterative fashion

Your answer is correct.

La risposta corretta è:
the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites

Domanda **16**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

The filter() method of DataFrame in Spark's structured API

- retrieves specified columns of a data frame for its rows which satisfy a boolean condition
- selects specified, possibly renamed, columns of a data frame
- accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition

Your answer is incorrect.

La risposta corretta è:
accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition

Domanda **17**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's structured API, calling the join method on a data frame df, with df1 as argument

- creates a new data frame containing all columns of df and df1; the computation is legal only if df and df1 have the same number of rows
- creates a new data frame containing all rows of df and df1; the computation is legal only if df and df1 have the same number of columns
- creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df

Your answer is correct.

La risposta corretta è:
creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df

Domanda **18**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The map function in MapReduce

- always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs
- is a predefined function which processes key-value pairs; its behaviour can be changed by passing parameters
- is a user defined function which is applied to every key-value pair of the dataset ✔

Your answer is correct.

Le risposte corrette sono:
always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs,

is a user defined function which is applied to every key-value pair of the dataset

Domanda **19**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the Hadoop system, the MapReduce application client

- centrally computes a map function, then negotiates resource allocation with YARN for the Reduce phase
- autonomously computes the resources each worker is allowed to use
- sends requests to YARN for resource allocation ✔

Your answer is correct.

La risposta corretta è:
sends requests to YARN for resource allocation

Domanda **20**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

A difference between an RDBMS and MapReduce is

- that an RDBMS is capable of performing distributed computation
- that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses ✔
- that MapReduce cannot compute any form of aggregation query on the data

Your answer is correct.

La risposta corretta è:
that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses

Domanda **21**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the structured API of Spark, pyspark.sql.Row

- is method of DataFrameReader which gets one row at a time from a JSON text file
- can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame ✔
- is method for reading rows from a data frame

Your answer is correct.

La risposta corretta è:
can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame

Vai a...

Quiz to Data Mining, Text Mining and Big Data Analytics - Theory [copy] ►

Iniziato	Wednesday, 21 December 2022, 12:17
Stato	Completato
Terminato	Wednesday, 21 December 2022, 12:53
Tempo impiegato	36 min. 12 secondi
Valutazione	Non ancora valutato

Domanda **1**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Which of the activities below is part of "Business Understanding" in the CRISP methodology?

- a. Which machine learning functions are necessary for my problem?
- b. Which data are available?
- c. Which data must be collected with a specific campaign?
- d. Which are the resources available (manpower, hardware, software, ...) ✔

Your answer is correct.

La risposta corretta è:
Which are the resources available (manpower, hardware, software, ...)

Domanda **2**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Which of the following are typical data warehouse queries?

Scegli una o più alternative:

- [x] a. Which products maximize the profit? ✔
- [x] b. What is the total revenue per product category and state? ✔
- [ ] c. What is the total revenue of yesterday in shop "BO03" ?
- [ ] d. Which is the stock of product "XXX" in the "YYY" warehouse
- [x] e. What is the relationship between profits gained by products "WWW" and "ZZZ"? ✔

Risposta corretta.

Le risposte corrette sono: Which products maximize the profit?, What is the total revenue per product category and state?, What is the relationship between profits gained by products "WWW" and "ZZZ"?

Domanda **3**
Risposta corretta
Punteggio ottenuto 1,00

Which of the texts describes the DFM schema below?

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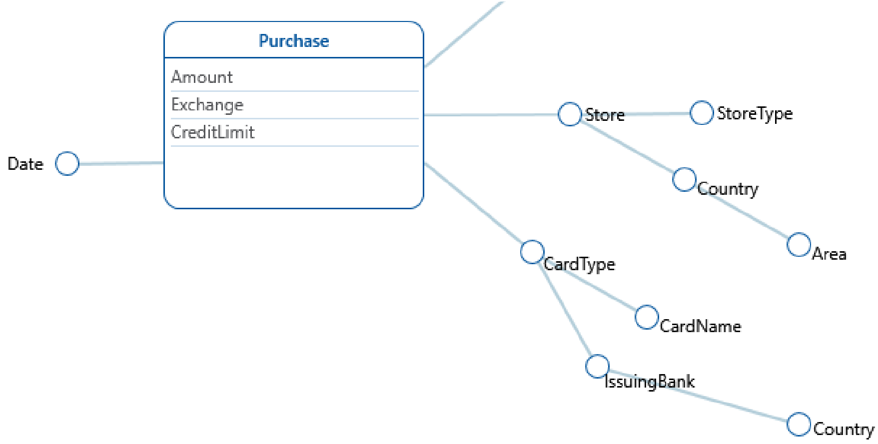
su 1,00



Scegli un'alternativa:

- a. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country.
- b. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores types have also a country and an area. Purchases have a card type (e.g. credit or debit) and a card name (e.g. MyCard or JeansCard). Card names have also an issuing bank and a country.
- c. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type and country; stores have also an area. Purchases have a card name (e.g. MyCard or JeansCard), card_names have a card type (e.g. credit or debit). Card types have also an issuing bank and a country.

Risposta corretta.

La risposta corretta è: We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country.

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about ETL, which of the following activities is related to the **Cleansing** step?

Scegli una o più alternative:

- a. **Snapshot** of the operational data
- b. Association of a **timestamp** to the operational data
- c. Elimination of **duplicates**
- d. Usage of dictionaries to solve **inconsistencies**

Risposta corretta.

Le risposte corrette sono: Elimination of **duplicates**, Usage of dictionaries to solve **inconsistencies**

Domanda **5**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the definition below describes the OLAP operation **Roll-Up**?

Scegli un'alternativa:

- a. Creates a link between concepts in interrelated cubes, to compare them
- b. Causes an increase in data aggregation and removes a detail level in a hierarchy ✔
- c. Reduces data aggregation and adds a detail level to a hierarchy
- d. Reduces the number of cube dimensions after setting one of the dimensions to a specific value
- e. Changes the layout, in order to analyse a group of data from a different viewpoint

Risposta corretta.

La risposta corretta è: Causes an increase in data aggregation and removes a detail level in a hierarchy

Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about the general idea of database, what is the meaning of "Schema on read"?

Scegli un'alternativa:

- a. Write the raw data in the database, then at reading time shape then according to the reader's needs
- b. Create a schema for data after reading them from the database ✔
- c. Change the schema of the data before each read
- d. Read the schema of the data before each data read

Risposta corretta.

Le risposte corrette sono: Write the raw data in the database, then at reading time shape then according to the reader's needs, Create a schema for data after reading them from the database

Domanda **7**

Completo

Punteggio max.: 1,00

Describe the concept of **Data Lake** and the main differences with respect to a *Data Warehouse* (maximum 1000 characters, the evaluation is graded according to clarity, completeness and relevance)

A data lake is a centralized repository for both structured and unstructured data. It is used to ingest huge quantities of data in a easy way and quickly, for this reason there is no need for ETL as happens for DWH. It can store data of various types and differently from the DW it has a schema on read and not a schema on write. It allows to store the majority of data of a company and index them with a catalog, so they can be retrieved easily when needed (this differences them from data swamps).

A user of a Data Lake can be both some business users as happens for DWH but also data scientist and this depends on the type of analyses that have to be done. Analyses can be of different kinds such as interactive or real time.

A Data Lake must have a low cost scalability and it must be easily scalable so that the costs for upgrading it are low. Differently from a DWH it must be cheap both in terms of time and costs to set up a Data Lake, which can be set up in a few days.

Two different architectures of data lakes are Lambda Lake and Kappa Lake.

Domanda **8**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Stemming and Lemmatization

- a. are text processing techniques to cut suffixes and reduce each word to its root and base form respectively ✔
- b. are text processing techniques to cut suffixes and reduce each word to its base form and root respectively
- c. are text processing techniques to cut prefix and reduce each word to its base form and root respectively
- d. are text processing techniques to cut prefix and reduce each word to its root and base form respectively

Your answer is correct.

La risposta corretta è:
are text processing techniques to cut suffixes and reduce each word to its root and base form respectively

Domanda **9**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Breakeven Point of Precision-Recall is

- a. an interpolated value of a precision-recall graph where precision and recall are identical ✔
- b. an interpolated value of a precision-recall graph where R-precision and recall are identical
- c. a value of a precision-recall graph where precision and recall are identical
- d. a value of a precision-recall graph where R-precision and recall are identical

Your answer is correct.

La risposta corretta è:
an interpolated value of a precision-recall graph where precision and recall are identical

Domanda **10**

Risposta errata

Punteggio

In a Chi-square test for feature selection of a term t with

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su 1,00

respect to a class c, the p-value is 0.015

- a. the term t is not selected as a feature if the confidence level is 0.99
- b. the term t is selected as a feature if the confidence level is 0.99 ✖
- c. the term t is not selected as a feature if the confidence level is 0.95
- d. the term t is selected as a feature if the confidence level is larger than 0.99

Your answer is incorrect.

La risposta corretta è:
the term t is not selected as a feature if the confidence level is 0.99

Domanda **11**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

a. 1. onehot encoding vector of q with respect to the dictionary ✔
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

b. 1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

c. 1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

d.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

Your answer is correct.

La risposta corretta è:
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

Domanda **12**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

Which is the correct sequence of layers in a typical Transformer Encoder (without considering residual connections and normalization layers) ?

a. Input embeddings -> Position Embeddings -> MultiHead Attention -> Feed Forward ✔

b. Feed Forward -> Input embeddings -> Position Embeddings -> MultiHead Attention

c. Position Embeddings -> Input embeddings -> MultiHead Attention -> Feed Forward

d. Input embeddings -> Position Embeddings -> Feed Forward -> MultiHead Attention

Your answer is correct.

La risposta corretta è:

Input embeddings -> Position Embeddings -> MultiHead Attention -> Feed Forward

Domanda **13**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

A big e-commerce company wants an information retrieval system to search for product images by providing a text query in natural language. They want their system to be as fast as

possible. Which of the following is a sensible approach to address this task ?

a. We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

b. We train a text encoder and an image encoder with metric learning to have a common latent space for text and images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird)

c. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

d. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird)

Your answer is correct.

La risposta corretta è:

We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

Domanda **14**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

Which are the components of a general message-passing layer in a graph neural network?

a. Message function, Permutation Invariant Aggregator, Differentiable Function

b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator

c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator

d. Message function, Permutation Invariant Aggregator, Attention Mechanism

d. Message function, Permutation Invariant Aggregator, Attention Mechanism

Your answer is incorrect.

La risposta corretta è:

Message function, Permutation Invariant Aggregator, Differentiable Function

Domanda **15**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The phrase "Big Data Loop" refers to

- a kind of big data processing framework which processes big data using a sequence of phases
- the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites ✔
- an algorithm designed to process big data in an iterative fashion

Your answer is correct.

La risposta corretta è:
the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites

Domanda **16**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

The filter() method of DataFrame in Spark's structured API

- retrieves specified columns of a data frame for its rows which satisfy a boolean condition ✘
- selects specified, possibly renamed, columns of a data frame
- accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition

Your answer is incorrect.

La risposta corretta è:
accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition

Domanda **17**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's structured API, calling the join method on a data frame df, with df1 as argument

- creates a new data frame containing all columns of df and df1; the computation is legal only if df and df1 have the same number of rows
- creates a new data frame containing all rows of df and df1; the computation is legal only if df

and df1 have the same number of columns

- creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df ✔

Your answer is correct.

La risposta corretta è:
creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df

Domanda **18**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

The map function in MapReduce

- always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs
- is a predefined function which processes key-value pairs; its behaviour can be changed by passing parameters
- is a user defined function which is applied to every key-value pair of the dataset ✔

Your answer is correct.

Le risposte corrette sono:
always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs,

is a user defined function which is applied to every key-value pair of the dataset

Domanda **19**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

In the Hadoop system, the MapReduce application client

- centrally computes a map function, then negotiates resource allocation with YARN for the Reduce phase
- autonomously computes the resources each worker is allowed to use
- sends requests to YARN for resource allocation ✔

Your answer is correct.

La risposta corretta è:
sends requests to YARN for resource allocation

Domanda **20**
Risposta corretta
Punteggio ottenuto 1,00 su 1,00

A difference between an RDBMS and MapReduce is

- that an RDBMS is capable of performing distributed computation
- that an RDBMS is optimised for workloads that are very different from the typical ones in ✔

Iniziato	Wednesday, 21 December 2022, 12:19
Stato	Completato
Terminato	Wednesday, 21 December 2022, 12:49
Tempo impiegato	29 min. 26 secondi
Valutazione	Non ancora valutato

Domanda **1**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In which part of the CRISP methodology we perform the **test design** activity?

- a. Modeling ✔
- b. Evaluation
- c. Business Understanding
- d. Data Understanding

Your answer is correct.

La risposta corretta è:
Modeling

Domanda **2**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following sentences describes an advantage of a Data Warehouse with respect to a standard DBMS

Scegli una o più alternative:

- [x] a. Allows analysis along the time dimension ✔
- [ ] b. Allows efficient execution of key-based queries
- [x] c. Allows efficient execution of multi-dimensional queries ✔
- [x] d. Has tools for helping to solve inconsistencies ✔
- [ ] e. Manages efficiently data updates

Risposta corretta.

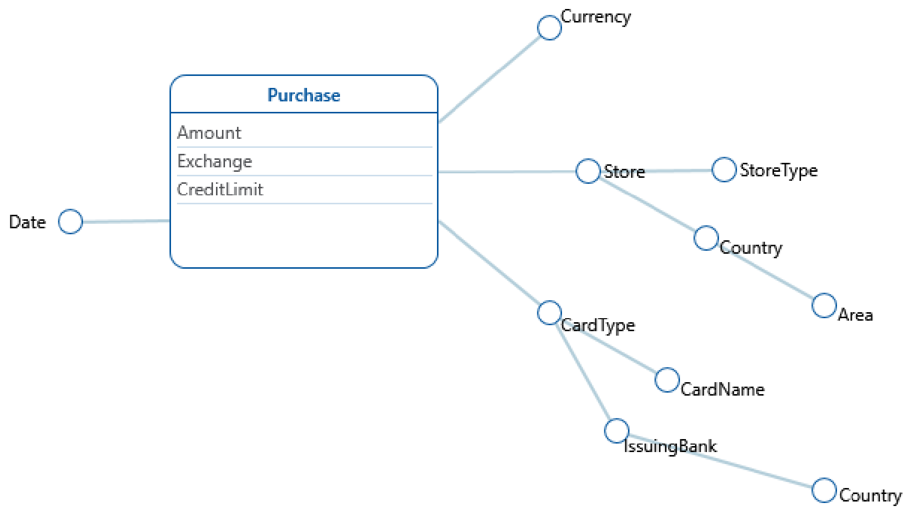
Le risposte corrette sono: Allows analysis along the time dimension, Allows efficient execution of multi-dimensional queries, Has tools for helping to solve inconsistencies

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the texts describes the DFM schema below?



Scegli un'alternativa:

- a. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country. ✔
- b. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores types have also a country and an area. Purchases have a card type (e.g. credit or debit) and a card name (e.g. MyCard or JeansCard). Card names have also an issuing bank and a country.
- c. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type and country; stores have also an area. Purchases have a card name (e.g. MyCard or JeansCard), card_names have a card type (e.g. credit or debit). Card types have also an issuing bank and a country.

Risposta corretta.

La risposta corretta è: We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country.

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about ETL, which of the following activities is related to the **Cleansing** step?

Scegli una o più alternative:

- a. **Snapshot** of the operational data
- b. Association of a **timestamp** to the operational data
- c. Elimination of **duplicates** ✔
- d. Usage of dictionaries to solve **inconsistencies** ✔

Risposta corretta.

Le risposte corrette sono: Elimination of **duplicates**, Usage of dictionaries to solve **inconsistencies**

Domanda **5**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the definition below describes the OLAP operation **Pivot**?

Scegli un'alternativa:

- a. Creates a link between concepts in interrelated cubes, to compare them
- b. Causes an increase in data aggregation and removes a detail level in a hierarchy
- c. Reduces data aggregation and adds a detail level to a hierarchy
- d. Reduces the number of cube dimensions after setting one of the dimensions to a specific value
- e. Changes the layout, in order to analyse a group of data from a different viewpoint ✔

Risposta corretta.

La risposta corretta è: Changes the layout, in order to analyse a group of data from a different viewpoint

Domanda **6**

Parzialmente corretta

Punteggio ottenuto 0,50 su 1,00

Talking about the general idea of database, what is the purpose of the "Schema on write" strategy?

Scegli una o più alternative:

- [x] a. Clean design of the data structure ✔
- [ ] b. Optimisation for specific types of queries
- [ ] c. Flexibility and efficiency for any kind of query
- [ ] d. Avoid preprocessing of data before writing

Risposta parzialmente esatta.

Hai selezionato correttamente 1.
Le risposte corrette sono: Clean design of the data structure, Optimisation for specific types of queries

Domanda **7**

Completo

Punteggio max.: 1,00

Describe the main principles and characteristics of a Data Warehouse (not more than 300 words)

DATA WArehouae is a type of decision support system.

IT HAS FEATURES SUCH AS:

- subject oriented

- non volatile: data is never overwritten or deleted.

- provides data in a uniform view

It can have diffrent structures:

- single layer: focuses on minimizing the space, no separation between operational and transactional process

- two layer: separation between transactional and analyttical process, adds a data staging layer

- three layer: has a reconciling layer

Domanda **8**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Why the TF-IDF of a stop word is zero ?

- a. because the IDF is 1
- b. because the TF is 1
- c. because the IDF is 0
- d. because the TF is 0

Your answer is correct.

La risposta corretta è:
because the IDF is 1

Domanda **9**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Breakeven Point of Precision-Recall is

- a. an interpolated value of a precision-recall graph where precision and recall are identical
- b. an interpolated value of a precision-recall graph where R-precision and recall are identical
- c. a value of a precision-recall graph where precision and recall are identical
- d. a value of a precision-recall graph where R-precision and recall are identical

Your answer is correct.

La risposta corretta è:
an interpolated value of a precision-recall graph where precision and recall are identical

Domanda **10**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

In a Chi-square test for feature selection of a term t with respect to a class c, the p-value is 0.015

- a. the term t is not selected as a feature if the confidence level is 0.99
- b. the term t is selected as a feature if the confidence level is 0.99
- c. the term t is not selected as a feature if the confidence level is 0.95
- d. the term t is selected as a feature if the confidence level is larger than 0.99 ✖

Your answer is incorrect.

La risposta corretta è:
the term t is not selected as a feature if the confidence level is 0.99

Domanda **11**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

a. 1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

b. 1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

c. 1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

d. 1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

Your answer is correct.

La risposta corretta è:
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which is the correct sequence of layers in a typical Transformer Encoder (without considering residual connections and normalization layers) ?

- a. Input embeddings -> Position Embeddings -> MultiHead Attention -> Feed Forward ✔
- b. Feed Forward -> Input embeddings -> Position Embeddings -> MultiHead Attention
- c. Position Embeddings -> Input embeddings -> MultiHead Attention -> Feed Forward
- d. Input embeddings -> Position Embeddings -> Feed Forward -> MultiHead Attention

Your answer is correct.

La risposta corretta è:

Input embeddings -> Position Embeddings -> MultiHead Attention -> Feed Forward

Domanda **13**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

A big e-commerce company wants an information retrieval system to search for product images by providing a text query in natural language. They want their system to be as fast as possible. Which of the following is a sensible approach to address this task ?

a. We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

b. We train a text encoder and an image encoder with metric learning to have a common latent space for text and images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird)

c. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

d. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird) ✖

Your answer is incorrect.

La risposta corretta è:

We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

Domanda **14**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which are the components of a general message-passing layer in a graph neural network?

- a. Message function, Permutation Invariant Aggregator, Differentiable Function
- b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator
- c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator
- d. Message function, Permutation Invariant Aggregator, Attention Mechanism

Your answer is correct.

La risposta corretta è:

Message function, Permutation Invariant Aggregator, Differentiable Function

Domanda **15**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

The phrase "Big Data Loop" refers to

- a kind of big data processing framework which processes big data using a sequence of phases ✖
- the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites
- an algorithm designed to process big data in an iterative fashion

Your answer is incorrect.

La risposta corretta è:
the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites

Domanda **16**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The filter() method of DataFrame in Spark's structured API

- retrieves specified columns of a data frame for its rows which satisfy a boolean condition
- selects specified, possibly renamed, columns of a data frame
- accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition ✔

Your answer is correct.

La risposta corretta è:
accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition

Domanda **17**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's structured API, calling the join method on a data frame df, with df1 as argument

- creates a new data frame containing all columns of df and df1; the computation is legal only if df and df1 have the same number of rows
- creates a new data frame containing all rows of df and df1; the computation is legal only if df and df1 have the same number of columns
- creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df ✔

Your answer is correct.

La risposta corretta è:
creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df

Domanda **18**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The map function in MapReduce

- always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs ✔
- is a predefined function which processes key-value pairs; its behaviour can be changed by passing parameters
- is a user defined function which is applied to every key-value pair of the dataset

Your answer is correct.

Le risposte corrette sono:
always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs,

is a user defined function which is applied to every key-value pair of the dataset

Domanda **19**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the Hadoop system, the MapReduce application client

- centrally computes a map function, then negotiates resource allocation with YARN for the Reduce phase
- autonomously computes the resources each worker is allowed to use
- sends requests to YARN for resource allocation ✔

Your answer is correct.

La risposta corretta è:
sends requests to YARN for resource allocation

Domanda **20**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

A difference between an RDBMS and MapReduce is

- that an RDBMS is capable of performing distributed computation
- that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses ✔
- that MapReduce cannot compute any form of aggregation query on the data

Your answer is correct.

La risposta corretta è:
that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses

Domanda **21**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the structured API of Spark, pyspark.sql.Row

- is method of DataFrameReader which gets one row at a time from a JSON text file
- can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame ✔
- is method for reading rows from a data frame

Your answer is correct.

La risposta corretta è:
can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame

Vai a...

Quiz to Data Mining, Text Mining and Big Data Analytics - Theory [copy] ►

Iniziato	Wednesday, 21 December 2022, 12:18
Stato	Completato
Terminato	Wednesday, 21 December 2022, 13:00
Tempo impiegato	42 min. 2 secondi
Valutazione	Non ancora valutato

Domanda **1**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In which part of the CRISP methodology we perform the **test design** activity?

a. Modeling ✔

b. Evaluation

c. Business Understanding

d. Data Understanding

Your answer is correct.

La risposta corretta è:
Modeling

Domanda **2**

Parzialmente corretta

Punteggio ottenuto 0,67 su 1,00

Select the sentences describing a characteristic of a *Data Warehouse*

Scegli una o più alternative:

- [ ] a. Non volatile
- [x] b. Constantly updated as soon as the base data are updated ✖ No, in general the DW is updated with a pre-defined frequency, according to the application needs
- [ ] c. Includes all the base data in their native format
- [x] d. Includes the time dimension ✔
- [x] e. Solves the inconsistencies ✔

Risposta parzialmente esatta.

Hai selezionato correttamente 2.

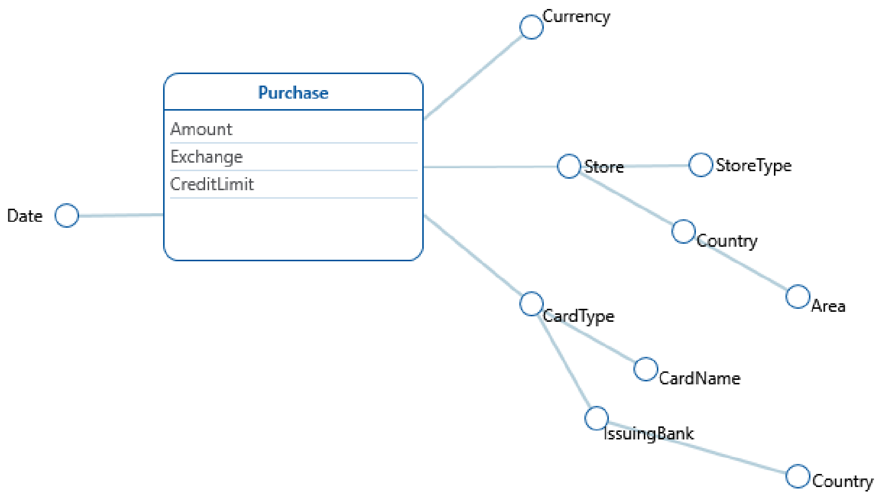
Le risposte corrette sono: Non volatile, Includes the time dimension, Solves the inconsistencies

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the texts describes the DFM schema below?



Scegli un'alternativa:

- a. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country. ✔
- b. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores types have also a country and an area. Purchases have a card type (e.g. credit or debit) and a card name (e.g. MyCard or JeansCard). Card names have also an issuing bank and a country.
- c. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type and country; stores have also an area. Purchases have a card name (e.g. MyCard or JeansCard), card_names have a card type (e.g. credit or debit). Card types have also an issuing bank and a country.

Risposta corretta.

La risposta corretta è: We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country.

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about ETL, which of the following activities is related to the **Extraction** step?

Scegli una o più alternative:

- [x] a. **Snapshot** of the operational data ✔
- [x] b. Association of a **timestamp** to the operational data ✔
- [ ] c. Elimination of **duplicates**
- [ ] d. Usage of dictionaries to solve **inconsistencies**

Risposta corretta.

Le risposte corrette sono: **Snapshot** of the operational data, Association of a **timestamp** to the operational data

Domanda **5**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Link the names of the OLAP operations below to their definitions

Roll-Up	Causes an increase in data aggregation and removes a detail level in a hierarchy	✔
Drill down	Reduces data aggregation and adds a detail level to a hierarchy	✔
Slice and dice	Reduces the number of cube dimensions after setting one of the dimensions to a specific value	✔
Pivot	Changes the layout, in order to analyse a group of data from a different viewpoint	✔
Drill across	Creates a link between concepts in interrelated cubes, in order to compare them	✔

Risposta corretta.

La risposta corretta è: Roll-Up → Causes an increase in data aggregation and removes a detail level in a hierarchy, Drill down → Reduces data aggregation and adds a detail level to a hierarchy, Slice and dice → Reduces the number of cube dimensions after setting one of the dimensions to a specific value, Pivot → Changes the layout, in order to analyse a group of data from a different viewpoint, Drill across → Creates a link between concepts in interrelated cubes, in order to compare them

Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about the general idea of database, what is the meaning of "Schema on read"?

Scegli un'alternativa:

- a. Write the raw data in the database, then at reading time shape then according to the reader's needs
- b. Create a schema for data after reading them from the database ✔
- c. Change the schema of the data before each read
- d. Read the schema of the data before each data read

Risposta corretta.

Le risposte corrette sono: Write the raw data in the database, then at reading time shape then according to the reader's needs, Create a schema for data after reading them from the database

Domanda **7**

Completo

Punteggio max.: 1,00

Describe the main objectives and benefits of the CRISP methodology

The CRISP methodology defines general guidelines which should be followed during a Data Mining Project.

Similarily to other Software Engineering methodologies, it's divided in structured phases which can be run one after the other through a loop which guarantees that the application performance can be updated or improved over time.

Each phase outputs different reports which can be used to track how dthe Data Mining Project is evolving, could be used as a reference for future Analysis and make the project reproducible in a similar setting.

As a general rule a Data mining Project should start with de definition of clear Business Objectives which could aim at improving a Company revenue or solving an Industrial/Social problem which could serve either as a product.

The business plan should be then analysed from a data point of view, since Gathering Data and processing it could be a cost intensive task (most likely the data is either external or should be heavy processed because it wasn't gathered for Data Mining purposes).

After Dathering the Data it should be used as an input for a Machine Learning model of any kind, used to tackle de problem defined during the first phase.

This model should then be evaluated to test performance with new data and put into production so that it can be used by employees and managers in the organization.

As mentioned before, all theese phases can be run in a cycle to continuously output new models and refine business objectives, since a company's needs usually change over time. Having a structured methodology helps dealing with changes in the objectives and create records for new Data scientists/Business Analysts/Machine Learning Engineers to keep up with their job tasks.

Domanda **8**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

Why the TF-IDF of a stop word is zero ?

- a. because the IDF is 1
- b. because the TF is 1
- c. because the IDF is 0
- d. because the TF is 0

Your answer is incorrect.

La risposta corretta è:
because the IDF is 1

Domanda **9**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The R-precision of a query q measures

- a. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q
- b. the number of relevant documents for the query q in the first R retrieved docs, divided by the number of docs in the corpus
- c. the number of relevant documents for the query q in the first R retrieved docs, divided by R which is the number of docs retrieved
- d. the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for each q

Your answer is correct.

La risposta corretta è:
the number of relevant documents in the first R docs retrieved, divided by R which is the number of relevant docs in the corpus for q

Domanda **10**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In a Chi-square test for feature selection of a term t with respect to a class c, the p-value is 0.015

- a. the term t is not selected as a feature if the confidence level is 0.99
- b. the term t is selected as a feature if the confidence level is 0.99
- c. the term t is not selected as a feature if the confidence level is 0.95
- d. the term t is selected as a feature if the confidence level is larger than 0.99

Your answer is correct.

La risposta corretta è:
the term t is not selected as a feature if the confidence level is 0.99

Domanda **11**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

a.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

b. ✖
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

c.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

d.
1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

Your answer is incorrect.

La risposta corretta è:
1. onehot encoding vector of q with respect to the dictionary

2. term weighting of the onehot encoded vector q

3. fold-in of the term weighted vector q in the LSA space

4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues

5. return the top-k highest cosine similarity LSA vector terms

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following terms is responsible for the quadratic complexity of the attention mechanism used inside Transformers (i.e., Attention(Q, K, V) = softmax(QK^T / sqrt(d_k))V) ?

a. The product between Q and K^T

b. The product between softmax(QK^T) and V

c. The factor 1/sqrt(d_k)

d. The V matrix

Your answer is correct.

La risposta corretta è:

The product between Q and K^T

Domanda **13**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

A big e-commerce company wants an information retrieval system to search for product images by providing a text query in natural language. They want their system to be as fast as possible. Which of the following is a sensible approach to address this task ?

- a. We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)
- b. We train a text encoder and an image encoder with metric learning to have a common latent space for text and images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird) ✖
- c. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)
- d. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird)

Your answer is incorrect.

La risposta corretta è:

We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

Domanda **14**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which are the components of a general message-passing layer in a graph neural network?

- a. Message function, Permutation Invariant Aggregator, Differentiable Function
- b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator
- c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator
- d. Message function, Permutation Invariant Aggregator, Attention Mechanism

Your answer is correct.

La risposta corretta è:

Message function, Permutation Invariant Aggregator, Differentiable Function

Domanda **15**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The phrase "Big Data Loop" refers to

- a kind of big data processing framework which processes big data using a sequence of phases
- the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites
- an algorithm designed to process big data in an iterative fashion

Your answer is correct.

La risposta corretta è:
the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites

Domanda **16**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The filter() method of DataFrame in Spark's structured API

- ○ retrieves specified columns of a data frame for its rows which satisfy a boolean condition
- ○ selects specified, possibly renamed, columns of a data frame
- ◉ accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition ✔

Your answer is correct.

La risposta corretta è:
accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition

Domanda **17**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's structured API, calling the join method on a data frame df, with df1 as argument

- ○ creates a new data frame containing all columns of df and df1; the computation is legal only if df and df1 have the same number of rows
- ○ creates a new data frame containing all rows of df and df1; the computation is legal only if df and df1 have the same number of columns
- ◉ creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df ✔

Your answer is correct.

La risposta corretta è:
creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df

Domanda **18**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The map function in MapReduce

- always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs
- is a predefined function which processes key-value pairs; its behaviour can be changed by passing parameters
- is a user defined function which is applied to every key-value pair of the dataset ✔

Your answer is correct.

Le risposte corrette sono:
always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs,

is a user defined function which is applied to every key-value pair of the dataset

Domanda **19**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

In the Hadoop system, the MapReduce application client

- centrally computes a map function, then negotiates resource allocation with YARN for the Reduce phase ✖
- autonomously computes the resources each worker is allowed to use
- sends requests to YARN for resource allocation

Your answer is incorrect.

La risposta corretta è:
sends requests to YARN for resource allocation

Domanda **20**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

A difference between an RDBMS and MapReduce is

- that an RDBMS is capable of performing distributed computation
- that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses ✔
- that MapReduce cannot compute any form of aggregation query on the data

Your answer is correct.

La risposta corretta è:
that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses

Domanda **21**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the structured API of Spark, pyspark.sql.Row

- is method of DataFrameReader which gets one row at a time from a JSON text file
- can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame ✔
- is method for reading rows from a data frame

Your answer is correct.

La risposta corretta è:
can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame

Iniziato	Wednesday, 21 December 2022, 12:18
Stato	Completato
Terminato	Wednesday, 21 December 2022, 12:38
Tempo impiegato	20 min. 4 secondi
Valutazione	**17,00** su un massimo di 21,00 (**81**%)

Domanda **1**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the activities below is part of "Business Understanding" in the CRISP methodology?

- a. Which machine learning functions are necessary for my problem?
- b. Which data are available?
- c. Which data must be collected with a specific campaign?
- d. Which are the resources available (manpower, hardware, software, ...)

Your answer is correct.

La risposta corretta è:
Which are the resources available (manpower, hardware, software, ...)

Domanda **2**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Rank the technologies for increasing level of abstraction: 1 = generate the most specific information, ... 3 = higher level knowledge

1 - SQL queries on operational databases

2 - OLAP analysis on Data Mart

3 - Data Mining on operational databases

Risposta corretta.

This question refers to the so-called *Business Intelligence Pyramid*

La risposta corretta è:
Rank the technologies for increasing level of abstraction: 1 = generate the most specific information, ... 3 = higher level knowledge

1 - [SQL queries on operational databases]

2 - [OLAP analysis on Data Mart]

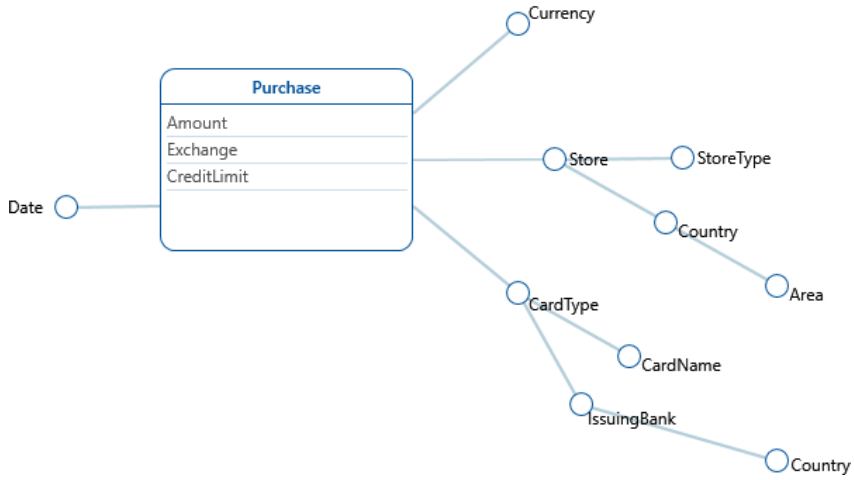
3 - [Data Mining on operational databases]

Domanda **3**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the texts describes the DFM schema below?



Scegli un'alternativa:

- a. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country. ✔
- b. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores types have also a country and an area. Purchases have a card type (e.g. credit or debit) and a card name (e.g. MyCard or JeansCard). Card names have also an issuing bank and a country.
- c. We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type and country; stores have also an area. Purchases have a card name (e.g. MyCard or JeansCard), card_names have a card type (e.g. credit or debit). Card types have also an issuing bank and a country.

Risposta corretta.

La risposta corretta è: We need to track the purchases with credit cards. Each purchase has an amount, an exchange and a credit limit. The purchases are done in a date and in a store of a given type; stores have also a country and an area. Purchases have a card type (e.g. credit or debit), card types have a card name (e.g. MyCard or JeansCard). Card types have also an issuing bank and a country.

Domanda **4**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about ETL, which of the following activities is related to the **Cleansing** step?

Scegli una o più alternative:

- a. **Snapshot** of the operational data
- b. Association of a **timestamp** to the operational data
- c. Elimination of **duplicates** ✔
- d. Usage of dictionaries to solve **inconsistencies** ✔

Risposta corretta.

Le risposte corrette sono: Elimination of **duplicates**, Usage of dictionaries to solve **inconsistencies**

Domanda **5**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the definition below describes the OLAP operation **Roll-Up**?

Scegli un'alternativa:

- a. Creates a link between concepts in interrelated cubes, to compare them
- b. Causes an increase in data aggregation and removes a detail level in a hierarchy ✔
- c. Reduces data aggregation and adds a detail level to a hierarchy
- d. Reduces the number of cube dimensions after setting one of the dimensions to a specific value
- e. Changes the layout, in order to analyse a group of data from a different viewpoint

Risposta corretta.

La risposta corretta è: Causes an increase in data aggregation and removes a detail level in a hierarchy

Domanda **6**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Talking about the general idea of database, what is the purpose of the "Schema on read" strategy?

Scegli una o più alternative:

- [x] a. Possibility to extract data in various shapes ✓
- [ ] b. Optimisation for various types of queries
- [x] c. Flexibility for any kind of query ✓
- [x] d. Avoid preprocessing of data before writing ✓

Risposta corretta.

Le risposte corrette sono: Possibility to extract data in various shapes, Flexibility for any kind of query, Avoid preprocessing of data before writing

Domanda **7**

Risposta non data

Punteggio max.: 1,00

Describe the concept of **Data Lake** and the main differences with respect to a *Data Warehouse* (maximum 1000 characters, the evaluation is graded according to clarity, completeness and relevance)

Domanda **8**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

Why the TF-IDF of a stop word is zero ?

- a. because the IDF is 1
- b. because the TF is 1
- c. because the IDF is 0 ✖
- d. because the TF is 0

Your answer is incorrect.

La risposta corretta è:
because the IDF is 1

Domanda **9**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Breakeven Point of Precision-Recall is

- a. an interpolated value of a precision-recall graph where precision and recall are identical ✔
- b. an interpolated value of a precision-recall graph where R-precision and recall are identical
- c. a value of a precision-recall graph where precision and recall are identical
- d. a value of a precision-recall graph where R-precision and recall are identical

Your answer is correct.

La risposta corretta è:
an interpolated value of a precision-recall graph where precision and recall are identical

Domanda **10**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In a Chi-square test for feature selection of a term t with respect to a class c, the p-value is 0.015

- a. the term t is not selected as a feature if the confidence level is 0.99 ✔
- b. the term t is selected as a feature if the confidence level is 0.99
- c. the term t is not selected as a feature if the confidence level is 0.95
- d. the term t is selected as a feature if the confidence level is larger than 0.99

Your answer is correct.

La risposta corretta è:
the term t is not selected as a feature if the confidence level is 0.99

Domanda **11**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

while the user is writing a query as a sequence of words in a search text field of a search engine, it proposes to the user a list of possible words to add to the query; for example the user has written the query "latent semantic" and before running the query the engine suggests some words to add to this query in descendant order of relevance such "analysis" or "indexing" or "retrieval" and so on.

Which are the steps to suggest the top-k terms to refine a query using an already created LSA space ?

- a. 1. onehot encoding vector of q with respect to the dictionary ✔

  2. term weighting of the onehot encoded vector q

  3. fold-in of the term weighted vector q in the LSA space

  4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues

  5. return the top-k highest cosine similarity LSA vector terms

b.

1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

c.

1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by their two eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

d.

1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector document, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector documents

Your answer is correct.

La risposta corretta è:

1. onehot encoding vector of q with respect to the dictionary
2. term weighting of the onehot encoded vector q
3. fold-in of the term weighted vector q in the LSA space
4. cosine similarity between the folded-in vector q and each LSA vector term, weighting the two vectors by all eigenvalues
5. return the top-k highest cosine similarity LSA vector terms

Domanda **12**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

Which of the following terms is responsible for the quadratic complexity of the attention mechanism used inside Transformers (i.e., Attention(Q, K, V) = softmax(QK^T / sqrt(d_k))V) ?

- a. The product between Q and K^T
- b. The product between softmax(QK^T) and V
- c. The factor 1/sqrt(d_k)
- d. The V matrix

Your answer is correct.

La risposta corretta è:

The product between Q and K^T

Domanda **13**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

A big e-commerce company wants an information retrieval system to search for product images by providing a text query in natural language. They want their system to be as fast as possible. Which of the following is a sensible approach to address this task ?

- a. We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)
- b. We train a text encoder and an image encoder with metric learning to have a common latent space for text and images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird)
- c. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)
- d. We train a text encoder and an image encoder with metric learning to have two distinct latent spaces: one for texts and one for images. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using efficient Transformers (e.g., Performer, BigBird) ✖

Your answer is incorrect.

La risposta corretta è:

We train a text encoder and an image encoder with metric learning to have a common latent space for text and image embeddings. To perform retrieval, we rank each image according to its distance from the query; this operation can be optimized using approximate k-NN strategies (e.g., Product Quantization, Inverted File Index)

Domanda **14**

Risposta errata

Punteggio ottenuto 0,00 su 1,00

Which are the components of a general message-passing layer in a graph neural network?

a. Message function, Permutation Invariant Aggregator, Differentiable Function

b. Message function, Neighborhood Sampling, Non-Permutation Invariant Aggregator

c. Message function, Neighborhood Sampling, Permutation Invariant Aggregator ✖

d. Message function, Permutation Invariant Aggregator, Attention Mechanism

Your answer is incorrect.

La risposta corretta è:

Message function, Permutation Invariant Aggregator, Differentiable Function

Domanda **15**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The phrase "Big Data Loop" refers to

- a kind of big data processing framework which processes big data using a sequence of phases
- the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites ✔
- an algorithm designed to process big data in an iterative fashion

Your answer is correct.

La risposta corretta è:
the generation of data as a consequence of the usage of commercial web site data by the owners of the web sites

Domanda **16**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The filter() method of DataFrame in Spark's structured API

- retrieves specified columns of a data frame for its rows which satisfy a boolean condition
- selects specified, possibly renamed, columns of a data frame
- accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition ✔

Your answer is correct.

La risposta corretta è:
accepts a string expression the meaning of which is similar to a SQL WHERE clause search condition

Domanda **17**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In Spark's structured API, calling the join method on a data frame df, with df1 as argument

- creates a new data frame containing all columns of df and df1; the computation is legal only if df and df1 have the same number of rows
- creates a new data frame containing all rows of df and df1; the computation is legal only if df and df1 have the same number of columns
- creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df ✔

Your answer is correct.

La risposta corretta è:
creates a new data frame containing all the columns of df and df1 and only rows such that their projection on the columns of df is a row of df

Domanda **18**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

The map function in MapReduce

- always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs
- is a predefined function which processes key-value pairs; its behaviour can be changed by passing parameters
- is a user defined function which is applied to every key-value pair of the dataset ✔

Your answer is correct.

Le risposte corrette sono:
always computes a random subset of the input key-value pairs first, then applies some user-defined transformation to the sampled pairs,

is a user defined function which is applied to every key-value pair of the dataset

Domanda **19**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the Hadoop system, the MapReduce application client

- centrally computes a map function, then negotiates resource allocation with YARN for the Reduce phase
- autonomously computes the resources each worker is allowed to use
- sends requests to YARN for resource allocation ✔

Your answer is correct.

La risposta corretta è:
sends requests to YARN for resource allocation

Domanda **20**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

A difference between an RDBMS and MapReduce is

- that an RDBMS is capable of performing distributed computation
- that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses ✔
- that MapReduce cannot compute any form of aggregation query on the data

Your answer is correct.

La risposta corretta è:
that an RDBMS is optimised for workloads that are very different from the typical ones in big data analyses

Domanda **21**

Risposta corretta

Punteggio ottenuto 1,00 su 1,00

In the structured API of Spark, pyspark.sql.Row

- is method of DataFrameReader which gets one row at a time from a JSON text file
- can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame ✔
- is method for reading rows from a data frame

Your answer is correct.

La risposta corretta è:
can be used to create an object which can be passed to a spark.createDataFrame method to add rows to a data frame

Vai a...

Quiz to Data Mining, Text Mining and Big Data Analytics - Theory [copy] ►